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Assessing the effectiveness of workplace accommodations in facilitating return-to-work after brain injury – a systematic review protocol.

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Assessing the effectiveness of workplace accommodations in facilitating return-to-work after brain injury – a systematic review protocol.

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Introduction: Returning to work and sustaining employment is a significant challenge for brain injury survivors. Within the literature, there is recurring support for the role of workplace accommodations in effective and early return-to-work (RTW). To date, however, there has been a lack of systematic reviews exploring the specific role of workplace accommodations within the context of RTW after brain injury. The primary objective of this protocol is to outline the methodological approach that will be undertaken to systematically review the literature and to assess the effectiveness of workplace accommodations in facilitating RTW.

Methods and Analysis: A total of nine databases will be searched systematically using the concepts 'Brain injury,' 'Return to work,' and 'Job Accommodations.' Grey Matters and the reference lists of eligible articles will also be searched. Study selection will be performed independently by three reviewers, based on predetermined eligibility criteria through two rounds of screening using, first, the title and abstract, followed by a full-text search. Extracted information will include the study's purpose, design, and setting; the data source and type; the severity of Traumatic Brain Injury (TBI) and the diagnostic criterion utilized; a comprehensive description of the intervention provided; the RTW outcome variables; and the statistical methods utilized etc. The data will be tabulated and narratively synthesized. Systematic review registration: This protocol has been registered with PROSPERO.

Ethics and Dissemination: As this review intends to use pre-existing published studies, ethical permissions will not be required. Nevertheless, this review will follow the ethical and governance standards in the data management and presentation of results. The findings from this review will potentially be published in a peer-reviewed scientific journal (electronically and in print). The results of this review will be presented at both national/international conferences and shared with stakeholders influencing return-to-work practices.

Keywords: Brain injury, Workplace accommodations, Return-to-work, Rehabilitation, Systematic review

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Strengths of this review:

- Significance is multi-fold; this focused review addresses the critical knowledge gap of how workplace accommodations can influence return-to-work outcomes and which category of accommodations have a greater impact on the sustenance of this return in the traumatic brain injury population has not been undertaken to date.
- Impact on health practitioners (physicians, rehabilitation professional, vocational counselors etc.) researchers, policy makers and survivors alike, i.e., acquisition of implementable knowledge regarding the nature of the accommodations and their impact; directing resources towards evidence-based and cost-effective solutions; engaging brain injury survivors to trial such accommodations early in their rehabilitation journey etc.
- Exploration using a sex/gender lens offers a distinct advantage to understand the sex- and gender-based influences on vocational outcomes.

Limitations:

- Heterogeneity of the traumatic brain injury population, employer/occupational variables and individual preferences may pose a challenge for more precise interpretations.
- The limited number of high level evidence within the vocational rehabilitation literature.

Implications:

Effective workplace accommodations promote an inclusive labor market by removing barriers. Utilizing the most effective workplace accommodations can provide employment opportunities to people with disabilities and vulnerabilities.

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BACKGROUND

Return-to-work (RTW) post-injury is a primary rehabilitation goal and perhaps the most reliable marker of functional recovery within a real-world context [1]. Returning to work and sustaining employment is a significant challenge for brain injury survivors [2]. It is estimated that 60% of working age individuals with moderate to severe acquired brain injury are unable to RTW post-injury [3] and a substantial proportion cannot sustain RTW. There is evidence that early and targeted vocational rehabilitation is cost-effective, and leads to improved work-related outcomes [2]. Since traumatic brain injury (TBI) more frequently occurs during prime working years, when people aim for vocational goals and are acquiring the skills needed for success [4], poor employment outcomes constitute a global health issue producing significant financial and societal burdens. The indirect economic cost of a TBI for work-aged Canadians is expected to increase to $8.2 billion by the year 2031, far exceeding costs of other common neurological conditions such as Spinal Cord Injury, Parkinson's Disease and Multiple Sclerosis [5].

TBI is caused by "a bump, blow, or jolt to the head or a penetrating head injury that disrupts the normal function of the brain" [6]. TBI is rated as mild, moderate and severe based on the initial trauma that caused the injury. Concussions and mild TBI (mTBI) are the most common of all TBIs [7], yet there is lack of understanding regarding their management in the workplace. mTBIs and concussions can have long-term functional effects on activities of daily living and return to productivity [8]. This is evidenced by the fact that the number of time loss claims for work-related concussions has increased by over 400% in Ontario, Canada, since 2005 [9]. The sequelea resulting from TBI are often diverse (i.e., headaches, memory difficulties, seizures, fatigue, visual and auditory deficits etc.) and accommodating for these impairments is necessary to ensure successful functioning in any re-integrative setting, including academic environments [10,11]. While memory loss and attention/concentration concerns are the most common functional limitations reported by TBI survivors, the persistent difficulties with emotional deregulation and the presence of medical symptoms have been identified as strong indicators of job retention [12].

Workplace accommodations (WAs) refer to the individually-tailored supports resulting from adaptations, alterations and/or modifications to job demands or workplace operations, which enable employee's equal access to essential aspects of their work [13, 14]. Within the context of personal injury, WAs attempt to facilitate re-integration to pre-injury work status [15]. WAs can be implemented in various forms across all stages of recovery to facilitate RTW. They may include provisions for flex days, opportunities for untimed breaks, the option to work from home, the arrangement of a job coach for specific work-oriented tasks etc [16,17].

Within the literature, there is recurring support for the role of WAs in effective and early RTW (i.e., shorter work absence duration [18], reduced disability costs [19], postponement or prevention of job loss, fewer job disruptions [20] and reduced productivity loss [21]). It is estimated that a significant proportion (83.5%) of workers with mental disorders need accommodations, yet only 30.5% receive them [22]. Workers who accept WAs are not only able to RTW but are also able to sustain work, compared to employees who rejected or did not receive an offer to accommodate [23]. In general, the use of compensatory strategies and supported employment practices enhance the likelihood of employment success [24]. Colantonio and colleagues [25] have shown that 76% of brain injury survivors have reported 'job modifications' and 'employers who accommodated treatment needs' as significant factors that facilitated RTW. More recently, Stergiou-Kita and colleagues have indicated that there is a lack of adequate understanding regarding the scope of WAs amongst both stakeholders [26,27]. A

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systematic review [28] did not find strong evidence to support that the patients' pre- or post-injury characteristics, or rehabilitation interventions has any effect on productivity status and vocational outcomes after TBI. Yet, the review did not consider organizational or vocational modifications such as accommodations and natural supports. Kristman and Colleagues [29], however, later reported that organizational factors are more important than demographic and occupational factors when considering RTW. Non-traditional determinants, such as family support, have been shown to play an equally important role in facilitating community and work re-integration [30].

There is evidence to suggest that sex/gender exerts significant influences on RTW processes, yet the influence of sex/gender on workplace injuries and outcomes is understudied [31,32]. Recent evidence reveals low and inconsistent support for the argument that gender significantly informs RTW after TBI [33, 34, 35], nevertheless positive outcome related to females in resuming work status after severe TBI is present [36]. After adjusting for demographic and health characteristics, adverse life events independently predicted increased rates of sickness-related absences among women [37]. It has been shown that while pink-collar employees are significantly more likely to receive an offer for accommodation, they are also more likely to refuse accommodations compared to their blue-collared counterparts [38]. The reasons for this discrepancy remain to be explored. A systematic review [39] on prognostic indictors of mTBI outcomes found that only 7% of >200 studies reviewed provided data stratified by sex. We are not aware of any other recent systematic reviews on RTW after TBI wherein the evidence was stratified by sex/gender.

To date, there is a lack of systematic reviews exploring the specific role of WAs within the context of RTW after TBI. While there are ample reviews that have investigated the role of work-related interventions on RTW, these reviews have focused on care and therapeutic programs or multidisciplinary intervention(s) that impact RTW including vocational rehabilitation. A more recent systematic review [40] found strong evidence that interventions with a combination of work-directed components are facilitating RTW in brain injury survivors. Nevertheless, the extracted data in this particular review paired rehabilitation interventions with work-related supports/aids to explain combined effects, versus stratifying for specific outcomes. Another scoping review concluded that WAs appear to be important for supporting employees with mental illness, but identified the implementation and evaluation of WAs as an area for further research [41]. Hence, while it is important to understand WAs from policy, legislative and social perspectives, it is perhaps more important to investigate the direct impact of WAs on employment re-entry and retention. Also, it is important to investigate the influence of sex/gender on the offer, acceptance and implementation of WAs.

The primary purpose of this protocol is to outline the methodology for a project that will systematically review the literature to identify the types of WAs that best facilitates re-entry to work and investigate the translational effect of successful WAs on the effectiveness of vocational outcomes, specifically RTW and, more importantly, the long-term sustenance of this return. The secondary aim is to consolidate and synthesize the quantitative and qualitative evidence in the literature. To achieve the latter goal, a mixed methods data synthesis approach will be undertaken to interpret the evidence. This review's final goal is to determine whether sex/gender variables are associated with the effectiveness of accommodations in promoting re-entry into the workplace.

METHODS/DESIGN

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PROSPERO

This protocol has been registered with PROSPERO (PROSPERO 2016:CRD42016043517)

PATIENT AND PUBLIC INVOLVEMENT

No patient involved.

SEARCH STRATEGY

The search strategy was developed with an information specialist at the Toronto Rehabilitation Institute-University Health Network (TRI-UHN). A modified PICO framework was used to facilitate and refine the search strategy. The search strategy was developed using a combination of comprehensive and specific subject headings and keywords, to balance sensitivity with specificity, and render articles across a wide spectrum of specific literature. Broader terms such as 'Return to Work', "Employment" and "Workplace" were used to search for articles, while specific narrow-focussed keywords such as "absenteeism", and "work" with "ability" and "capacity" were used to search relevant non-traditional publications. Similarly, generic heading such as "equipment and supplies" and "job accommodations" were searched alongside alternatives including "jobsite" with "interventions" and "duties." These alternative terms helped capture publications that may exclude more commonly deployed workplace adaptation keywords. Search terms were also adapted from previously published systematic reviews on the topic [42,43].

The search terms and strategy were independently peer-reviewed by another information specialist. Changes and suggestion were incorporated before the search terms were translated across nine databases, i.e., 1) Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations, 2) EBM Reviews Cochrane Central Register of Controlled Trials, 3) EBM Reviews Cochrane Database of Systematic Reviews, 4) Embase, 5) PsycINFO, 6) CINAHL, 7) Pubmed, 8) OT SEEKER, and 9) Health and Safety Science Abstracts. Additional supplementary file provides the search strategy for each database. Furthermore, conference materials such as abstracts, editorials and/or commentaries will be excluded as they do not provide sufficient information required for the purposes of this review. All database results will be limited to English language. A limit to human only studies will be applied.

The grey literature will be searched for additional information using both 'Grey Matters' (a practical search tool for evidence based medicine) and resources from known relevant rehabilitation research groups (i.e., Institute of Work and Health, Job Accommodations Network, Employment Accessibility Resource Network etc). The reference lists of all included articles will be hand searched.

STUDY SELECTION

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for systematic reviews will be used to report the findings of this review [44].

Study selection will be completed by three reviewers, pre-determined eligibility criteria will guide all three reviewers and screening will be undertaken in two steps. Level I screen will involve screening of studies based on their titles and abstracts only. The Level I screen inclusion criteria are: 1) study participants must have sustained a TBI (irrespective of the mechanism or place of injury), 2) an outcome related to RTW must have been reported, and 3) study participants must be of working age (18-65 years). The primary outcome of interest, 'return-to-work', is defined as a post-injury attempt to resume productive activities. Each article may have

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one of three possible rating, i.e., 'yes', 'no' and 'maybe'. All three reviewers will discuss articles that have non-matched ratings; if consensus is not achieved, mediation consultation will be undertaken. Articles that pass the first screen criteria will proceed to the full text Level II screen. The Level II screen entails a full-text screening of articles. Two additional criteria will be considered during the full-text Level II screen: 1) study participants were working at the time of the injury/loss, and 2) an intervention that includes WAs (of any kind) was provided to facilitate the RTW.

For the purposes of this review, our operational definition of 'accommodation' is "any measurable change and/or alteration in the process flow of an organization, material used to accomplish a job demand, services provided/utilized at the workplace and/or routines for oneself (or others) to compensate for impairments or modify a work environment after a condition, disorder, injury and/or illness in an attempt to facilitate re-entry into the workforce" [Adapted from [45, 46]]. All study designs will be included, for instance, randomized and non-randomized controlled trials, case-control studies, cohort and longitudinal studies, retrospective analysis, scoping or systematic reviews, case studies, qualitative studies and/or mixed methods research studies. Qualitative and case studies will particularly be included as they may contain details which quantitative exploration alone cannot capture. With respect to vocational status, all types of employment (full-time, part-time, casual, contract, self-employed, volunteer etc.) and occupational categories will be considered. The same will be true for injury severity, whereby all mild, moderate and severe injuries will be considered. With sufficient data, analysis will be stratified by sex, age and injury severity. Data from international publications, albeit written in English, will be included.

DATA EXTRACTION AND MANAGEMENT

All three reviewers will develop and pilot the data extraction form. The form will be modified to ensure that relevant data is captured for the assessment of study quality. The data extraction form will include author(s), study title, publication year, purpose of study, study design, study setting, sample size, demographic and vocational characteristics of the study participants, data source or type (e.g., retrospective, interviews etc), severity of TBI and the diagnostic criterion utilized, statistical methodology utilized (e.g., descriptive, analytic and/or stratification techniques), comprehensive description of the intervention provided (e.g., employer-initiated, RTW co-ordinator, occupational therapist, case manager etc), duration of RTW, subsequent follow-up(s), characteristics and costs of the accommodations provided, outcome of the stated intervention etc. If possible, secondary outcomes such as quality of life, health status and satisfaction indicators will be reported alongside stated conflicts of interest and 'Risk of Bias' assessments.

Study data included after the level II screen will be extracted by two reviewers using a blinded extraction process. Reviewers will cross-review the collected data to document and address any discrepancies and rationales. Following this comparative analysis, the revised data extraction will be peer-reviewed by the third reviewer. Completed data extraction forms will be archived, and information from these forms will be inputted and stored in Excel. One reviewer will independently check the data entry for accuracy.

QUALITY ASSESSMENT

The Critical Appraisal Skills Programme (CASP) checklists [47] will be used to assess study quality by guiding the assessment of validity and reliability of the included studies. The

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rationale for selecting the CASP checklists was made primarily due to the availability of separate checklists for different study designs such as cohort studies, review papers, randomized controlled trials and qualitative studies. Two reviewers will complete the checklists, through a blinded process, and later cross-review to document and address any discrepancies and rationales. Following consensus, the third reviewer will peer-review all checklists associated with this review's articles. With respect to missing data, study authors may be contacted for any queries regarding relevant data capture or additional data required to assess the risk of bias.

DATA ANALYSIS, SYNTHESIS AND REPORTING

Data will be collected from each study and tabulated in a summary table. Due to the anticipated heterogeneity of study designs, patient cohorts and variability of outcome measure, a meta-analytical approach may not be feasible. Nevertheless, descriptive statistics will be utilized to synthesize the evidence from quantitative studies while data from qualitative studies will be summarized by identifying prominent and prevalent themes. This approach is intended to enhance the utility and impact of the findings [48]. A data analysis strategy [49] may be used to aggregate results, integrate the original meaning of the primary studies and reconstruct the findings to facilitate new interpretations of the data. The sources of the synthesized information will be reported separately, however, attempts to mix and combine the results of both quantitative and qualitative data will be made to provide a comprehensive conceptual understanding of the findings.

Sub-group analysis: With sufficient data, sub-group analyses will be undertaken to investigate differences in outcomes. Given the expansion of the literature with respect to sex/gender differences in outcomes, two relevant Cochrane tools will be utilized, i.e., the Sex and Gender in Systematic Reviews Planning Tool and the Equity Checklist for Systematic Review Authors [50,51]. This unique investigation will clarify differences observed in data collection, analysis, and interpretation across studies. To examine the impact of interventions on inequality, the data will be further assessed using an intersectional lens to better understand the contributions of social identities (e.g., sex, age, race, socioeconomic status, etc.) as well as disadvantaged populations on the outcomes of RTW and productivity.

The review will be undertaken according to this published protocol. Any deviations from the protocol will be reported in a separate sub-section and labeled "Differences between published protocol and review."

DISCUSSION

We believe this focused undertaking is the first of its kind. At this time, there have been no systematic reviews exploring the effectiveness of specific WAs that facilitate and/or sustain RTW after TBI. Reviews published to date have considered the role of a combination of care programs such as outpatient rehabilitation programs, generic workplace interventions, the impact of community-based remediation on RTW and/or return-to-productivity outcomes.

The Convention on the Rights of Persons with Disabilities (CRPD) [52], Disability Discrimination Act (DDA) [53] as well as the Canadian Human Rights Act (CHRC) [14] mandates equal rights and opportunities at the workplace, i.e., employees with disabilities are entitled to the same opportunities and benefits as people without disabilities. Under the legal test, CHRC, the employer has a duty to take reasonable steps to accommodate the position of an employee unless the employer's business would suffer 'undue hardship'. Although the Guides for Assessing Persons with Disabilities [45] have provided some guidance regarding the interpretation of undue

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hardship, operationalizing the construct of WAs has been short of easy. We believe this review will help shed light on the various conceptualizations of accommodations within the workplace and how WAs are operationalized within the practice of RTW following TBI. Investigating effective interventions that facilitate returning TBI survivors to work not only has the potential for short-term impact on a person's working life and ability to contribute to the economy, but also on the long-term health and well-being of society.

Given the evolving workforce, certain work-related supports are naturally embedded within the workplace, available for all employees and not considered a customized adjustment. These are often referred to as 'natural supports'. It is important to distinguish between WAs and natural supports, the former being a support customized for a particular worker who is encountering difficulties with work resumption.

Effective WAs holds the potential to foster inclusive work designs for TBI survivors who are considering a return to employment. Given that brain injury occurs frequently among working age individuals, the study has implications from a labor market policy perspective. The information gathered and consolidated in this review will help health practitioners, policy makers, researchers and other workforce-related stakeholders make informed decisions when considering service delivery or future research into RTW for brain injury survivors. With respect to service delivery, the 'low cost and high impact'[54] feature of WAs makes it not only easy to implement but also economically lucritive. Added benefits of WAs include retention of previously trained employees and hence elimination of costs associated with training a new employee, boosted morale amongst workers and an increased sense of well-being at both individual and societal levels.

Traditional RTW practices focus excessively on sickness and disability, whereby, based on a reported mismatch between the client's limitations and the demands of the job - a worker is deemed "unfit to work". In 2010, the United Kingdom replaced its "sickness certificate" with a statement of fitness for work also known as the "fit note" [55]. The shift in methodology from "not fit to work" to "fit to work" (if the employer is able to make necessary modifications) transformed the traditional "disability" lens into a "functional" focus. Currently, many procedural, legislative, training and practical issues remain regarding the nature of the ideal fit note, nevertheless, there is evidence to support that these notes, when filled correctly, are not only useful to the employer for planning an employee's return but are also improving RTW rates by more than 80% [56]. Health Care Practitioner's, Physician's and employer's knowledge of effective WAs can facilitate work re-entry by ensuring an inclusive and safe work environment for the worker.

Also, the findings of this review are transferable across other sectors concerned with RTW or sustainability at work. We will provide a few examples for clarity's sake. First, the review aims to investigate the reported sex/gender based differences within the context of WAs and RTW. Since women, on average, have shorter earning careers and interruptions related to motherhood and unequal sharing of care/housework demands and early retirement [57,58], effective WAs will help them sustain employment longevity to support retirement/pension planning. Vocational longevity also helps facilitates continuous employment through various stages of a female's career (i.e., childbearing, life transitions) and combats the female wage inequality gap. Second, workplaces have to become more accommodating of older workers particularly owing to age-related cognitive decline. By 2031, approximately 25% of Canada's population is expected to be age 65 or older [59]. Moreover, Canadians are working longer and retiring later. Over the past few decades, the average retirement age has risen from 61 to 63 years

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[60]. Similar trends are observed in the United Kingdom [61]. Effective WAs not only lessen the economic burden associated with TBI but also create opportunities for the aging population by enabling their productivity post-retirement.

Effective WAs, for workers with special needs, fosters increased productivity which leads to economic well-being and eventually can be operationalized as the living standards for the population. Creating an inclusive labor market by removing barriers, and providing employment opportunities to people with disabilities, will not only generate a high impact return on the investment but also maximize the utility of human capital.

DISSEMINATION PLAN

In addition to traditional knowledge translation approaches (i.e., publishing the findings in peer-reviewed journals, presenting at local and international conferences, posting summary-of-findings on university websites etc), findings from this review will be disseminated using an integrated action-oriented knowledge translation framework [62]. It is hoped that this action-oriented framework will enable contextualization of the knowledge and findings to ensure impact in the real-world. Initial interest will be ignited through free webinars and podcasts as well as social media engagements, such as Twitter. Later, various stakeholders (i.e., Workplace Safety and Insurance Board, Insurance Bureau of Canada, Accessibilty Ontario etc.) will be invited to engage in a discussion to: 1) understand and contextualize the knowledge gained and 2) to explore how to reduce potential uptake barriers within the workplace environment. A special rehabilitation focus will be maintained to translate the knowledge to health practitioners, and to facilitate the development guidelines that will help facilitate the integration of these findings into day-to-day RTW practice.

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Contributors: AC and RS conceptualized the study. RS formulated the design, drafted the manuscript, assisted with the development of the search terms and registered the protocol via PROSPERO. AC provided expertise at each level and also reviewed the protocol. Both authors read and approved the final manuscript.

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References

1 Franulic, A., Carbonell, C. G., Pinto, P., & Sepulveda, I. (2004) Psychosocial adjustment and employment outcome 2, 5 and 10 years after TBI, Brain Injury, 18:2, 119-129

2 Radford, K., Phillips, J., Drummond, A., Sach, T., Walker, M., Tyerman, A., Haboubi, N., & Jones, T. (2013). Return to work after traumatic brain injury: cohort comparison and economic evaluation. Brain Inj, 27(5), 507-520.

3 van Velzen, J. M., van Bennekom, C. A., Edelaar, M. J., Sluiter, J. K., & Frings-Dresen, M. H. (2009). How many people return to work after acquired brain injury?: A systematic review, Brain Injury, 23:6, 473-488

4 Sarajuuri, J. M., Kaipio, M. L., Koskinen, S. K., Niemela, M. R., Servo, A. R., & Vilkki, J. S. (2005). Outcome of a comprehensive neurorehabilitation program for patients with traumatic brain injury. Arch Phys Med Rehabil, 86(12), 2296-2302.

5 Public Health Agency of Canada and Health Canada Canada. (2014). Mapping Connections: An Understanding of Neurological Conditions in Canada. Retrieved from: http://www.phac-aspc.gc.ca/publicat/cd-mc/mc-ec/index-eng.php

6 Centers for Disease Control and Prevention (2016). TBI: Get the Facts. Retrieved from: http://www.cdc.gov/traumaticbraininjury/get_the_facts.html

7 National Center for Injury Prevention and Control. Report to Congress on Mild Traumatic Brain Injury in the United States: Steps to Prevent a Serious Public Health Problem. Atlanta, GA: Centers for Disease Control and Prevention; 2003.

8 Cancelliere, C., Kristman, V. L., Cassidy, D., Hincapié, C. A., Côté, P., Boyle, E., Carroll, L. J., Stålnacke, B., Boussard, c. N., & Borg, J. (2014). Systematic Review of Return to Work After Mild Traumatic Brain Injury: Results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Archives of Physical Medicine and Rehabilitation 95(3): S201-S209.

9 Workplace Safety & Insurance Board (2015). By The Numbers: 2014 WSIB Statistical Report. Retrieved from: http://www.wsibstatistics.ca/

10 Jantz, P. B., & Coulter, G.A. (2007). Child and adolescent traumatic brain injury: Academic, behavioural, and social consequences in the classroom. Support for Learning, 22(2), 84-89.

11 Hendricks, D. J., Sampson, E., Rumrill, P., Leopold, A., Elias, E., Jacobs, K., Nardoned, A., Scherere, M., & Stauffer, C. (2015). Activities and interim outcomes of a multi-site development project to promote cognitive support technology use and employment success among postsecondary students with traumatic brain injuries. NeuroRehabilitation, 37(3), 449-458.

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12 Artman L. K., & McMahon B.T. (2013). Functional limitations in TBI and their relationship to job maintenance following work re-entry. Journal of Vocational Rehabilitation 39, 39, 13–21.

13 Americans With Disabilities Act [Pub. L. No. 101-336, 104 Stat. 328, 1990].

14 Canadian Human Rights Act [R.S.C., 1985, c. H-6].

15 Mueller, J. L. (1999). Returning to work through job accommodation: a case study. AAOHN Journal, 47(3), 120-129; quiz 120-131.

16 Targett, P., & Wehman, P. (2010) Return to work after traumatic brain injury: a supported employment approach. In: Schultz IZ, Rogers ES (eds) Handbook of work accommodatins and retention in mental health. Springer, New York

17 Schultz, I. Z., Winter, A., & Wald, J. (2010) Evidentiary support for best practices in job accommodation in mental health: macrosystem, employer-level and employee-level interventions. In: Schultz, I. Z., Rogers, E. S. (eds) Handbook of work accommodation and retention in mental health. Springer, New York

18 Franche, R. L., Cullen, K., Clarke, J., Irvin, E., Sinclair, S., Frank, J., & Institute for Work & Health (IWH) Workplace-Based RTW Intervention Literature Review Research Team. (2005). Workplace-based return-to-work interventions: a systematic review of the quantitative literature. J Occup Rehabil, 15(4), 607-631.

19 Franche, R. L., Severin, C. N., Hogg-Johnson, S., Cote, P., Vidmar, M., & Lee, H. (2007). The impact of early workplace-based return-to-work strategies on work absence duration: a 6-month longitudinal study following an occupational musculoskeletal injury. J Occup Environ Med, 49(9), 960-974.

20 Johnson, K. L., Yorkston, K. M., Klasner, E. R., Kuehn, C. M., Johnson, E., & Amtmann, D. (2004). The cost and benefits of employment: a qualitative study of experiences of persons with multiple sclerosis. Arch Phys Med Rehabil, 85(2), 201-209.

21 Gignac, M. A., Cao, X., & McAlpine, J. (2015). Availability, need for, and use of work accommodations and benefits: are they related to employment outcomes in people with arthritis? Arthritis Care Res (Hoboken), 67(6), 855-864.

22 Wang, J., Patten, S., Currie, S., Sareen, J., & Schmitz, N. (2011). Perceived needs for and use of workplace accommodations by individuals with a depressive and/or anxiety disorder. J Occup Environ Med, 53(11), 1268-1272.

23 Franche, R. L., Severin, C. N., Hogg-Johnson, S., Lee, H., Cote, P., & Krause, N. (2009). A multivariate analysis of factors associated with early offer and acceptance of a work

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accommodation following an occupational musculoskeletal injury. J Occup Environ Med, 51(8), 969-983.

24 Target, P., Wehman, P., Petersen, R., & Gorton, S. (1998). Enhancing work outcome for three persons with traumatic brain injury. Int J Rehabil Res, 21(1), 41-50.

25 Colantonio, A., & Lee, V. (2012). Return to work after mild to moderate work related injury: A pilot study. Brain Injury, 26 (4-5), 778.

26 Stergiou-Kita, M., Mansfield, E., & Colantonio, A. (2014). Injured workers' perspectives on how workplace accommodations are conceptualized and delivered following electrical injuries. J Occup Rehabil, 24(2), 173-188.

27 Stergiou-Kita, M., Daiter, L., & Colantonio, A. (2015). Good Intentions? Employer Representative Conceptualizations and, Challenges to the Workplace Accommodation Process: The Case of Electrical Injuries. Employ Respons Rights J, 27, 1–25.

28 Saltychev, M., Eskola, M., Tenovuo, O., & Laimi, K. (2013). Return to work after traumatic brain injury: Systematic review. Brain Inj, 27(13-14), 1516-1527.

29 Kristman, V. L., Shaw, W. S., Reguly, P., Williams-Whitt, K., Soklaridis, S., & Loisel, P. (2016). Supervisor and Organizational Factors Associated with Supervisor Support of Job Accommodations for Low Back Injured Workers. J Occup Rehabil.

30 Gagnon, A., Lin, J., & Stergiou-Kita, M. (2016). Family members facilitating community re-integration and return to productivity following traumatic brain injury - motivations, roles and challenges. Disabil Rehabil, 38(5), 433-441.

31 Stergiou-Kita, M., Mansfield, E., Sokoloff, S., & Colantonio, A. (2016). Gender Influences on Return to Work After Mild Traumatic Brain Injury. Archives of Physical Medicine and Rehabilitation, 97(2 Suppl), S40-45.

32 Colantonio, A. (2016). Sex, Gender, and Traumatic Brain Injury: A Commentary. Archives of Physical Medicine and Rehabilitation, Volume 97 , Issue 2 , S1 - S4

33 Keyser-Marcus, L. A., Bricout, J. C., Wehman, P., Campbell, L. R., Cifu, D. X., Englander, J., High, W., & Zafonte, R. D. (2002). Acute predictors of return to employment after traumatic brain injury: A longitudinal follow-up. Archives of Physical Medicine and Rehabilitation, 83(5), 635-641.

34 Kreutzer, J. S., Marwitz, J. H., Walker, W., Sander, A., Sherer, M., Bogner, J., Fraser, R., & Bushnik, T. (2003). Moderating factors in return to work and job stability after traumatic brain injury. J Head Trauma Rehabil, 18(2), 128-138.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

35 Ownsworth, T., & McKenna, K. (2004). Investigation of factors related to employment outcome following traumatic brain injury: a critical review and conceptual model. Disabil Rehabil, 26(13), 765-783.

36 Scherzer, F. C. P. (1993). Predictors and indicators of work status after traumatic brain injury: A meta-analysis. . Neuropsychological Rehabilitation, 3(1), 5-35.

37 Suominen, S., Vahtera, J., Korkeila, K., Helenius, H., Kivimaki, M., & Koskenvuo, M. (2007). Job strain, life events, and sickness absence: a longitudinal cohort study in a random population sample. J Occup Environ Med, 49(9), 990-996.

38 Franche, R. L., Severin, C. N., Hogg-Johnson, S., Lee, H., Cote, P., & Krause, N. (2009). A multivariate analysis of factors associated with early offer and acceptance of a work accommodation following an occupational musculoskeletal injury. J Occup Environ Med, 51(8), 969-983.

39 Cancelliere, C., Donovan, J., & Cassidy, J. D. (2016). Is Sex an Indicator of Prognosis After Mild Traumatic Brain Injury: A Systematic Analysis of the Findings of the World Health Organization Collaborating Centre Task Force on Mild Traumatic Brain Injury and the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch Phys Med Rehabil, 97(2 Suppl), S5-18.

40 Donker-Cools, B. H. P. M., Daams, J. G., Wind, H., & Frings-Dresen, M. H. W. (2016). Effective return-to-work interventions after acquired brain injury: A systematic review. Brain Injury, 30(2), 113-131.

41 McDowell, C., & Fossey, E. (2015). Workplace accommodations for people with mental illness: a scoping review. J Occup Rehabil, 25(1), 197-206.

42 Nieuwenhuijsen, K., Faber, B., Verbeek, J.H., Neumeyer-Gromen, A., Hees, H.L., Verhoeven, A.C., van der Feltz-Cornelis, C.M., and Bültmann, U.(2014). Interventions to improve return to work in depressed people. Cochrane Database Syst Rev, 3;(12): CD006237

43 van Vilsteren, M., van Oostrom, S. H., de Vet, H. C., Franche, R. L., Boot, C. R., and Anema, J.R. (2015). Workplace interventions to prevent work disability in workers on sick leave. Cochrane Database Syst Rev, 5(10):CD006955

44 Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., & Group, T.P. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.

45 Public Service Commission of Canada. (2007). Guide for Assessing Persons with Disabilities: How to determine and implement assessment accommodations. Retrieved from: http://www.psc-cfp.gc.ca/plcy-pltq/guides/assessment-evaluation/apwd-eph/pdf/apwd-eph-eng.pdf

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

46 Job Accommodations Network. Available at: http://askjan.org

47 Critical Appraisal Skills Programme (CASP) Checklists (2013). Retrieved from: http://www.casp-uk.net/casp-tools-checklists

48 Harden, A. (2010). Mixed-Methods Systematic Reviews: Integrating Quantitative and Qualitative Findings. Technical Brief no. 25.

49 Jang, E. E., McDougall, D. E., Herbert, M., Pollon, D., & Russell, P. (2008). Integrative mixed methods data analytic strategies in research on school success in challenging circumstances. Journal of Mixed Methods Research, 2, 221-247.

50 Doull M., Runnels V., Tudiver S., Boscoe M. (2011). Sex and gender in systematic reviews: Planning Tool.

51 Ueffing E, Tugwell P, Welch V, Petticrew M, Kristjansson E for the Campbell and Cochrane Equity Methods Group. Equity Checklist for Systematic Review Authors. Version 2012-10-02. Retrieved from: http://equity.cochrane.org/sites/equity.cochrane.org/files/uploads/EquityChecklist2012.pdf

52 United Nations General Assembly, Convention on the Rights of Persons with Disabilities [Resolution 61/106]. Retrieved from: http://www.ohchr.org

53 Disability Discrimination Act [1995, c. 50]. Retrieved from: http://www.legislation.gov.uk/ukpga/1995/50/contents

54 Office of Disability Employment Policy. (2007). Workplace Accommodations: Low Cost, High Impact. Retrieved from http://digitalcommons.ilr.cornell.edu/gladnetcollect/383

55 Department for Work and Pensions. (2013). Guidance for GPs, employers, hospital doctors and occupational health practitioners on using fit notes (statements of fitness for work).. Retrieved from: https://www.gov.uk/government/collections/fit-for-work-guidance

56 Coole, C., Potgieter, I., Nouri, F., Worthington, E., & Drummond, A. (2015). Return-to-work outcomes and usefulness of actual fit notes received by employers. Fam Pract, 32(5), 551-556.

57 Gower, D. (1997). "Measuring the age of retirement." Perspectives on Labour and Income: Statistics Canada, Catalogue no. 75-001-XPE, 9 (2)): 11-17.

58 Angelov, N., Johansson, P., and Lindahl, E. (2013). Is the Persistent Gender Gap in Income and Wages Due to Unequal Family Responsibilities?. IZA Discussion Paper No. 7181. Available at SSRN: https://ssrn.com/abstract=2210841

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59 Denton, F.T., Feaver, C.H. & Spencer, B.G. (2005). Time series analysis and stochastic forecasting: An econometric study of mortality and life expectancy. Journal of Population Economics18: 203-227

60 Statistics Canada. (2015). Study: The local unemployment rate and retirement, 1991 to 2007. Retrieved from: http://www.statcan.gc.ca/daily-quotidien/150422/dq150422a-eng.pdf

61 Office of National Statistics. (2013). Pension Trends – Chapter 4: The Labour Market and Retirement. Retrieved from: http://webarchive.nationalarchives.gov.uk/20160105160709/http://www.ons.gov.uk/ons/rel/pensions/pension-trends/chapter-4--the-labour-market-and-retirement--2013-edition/art-pt2013ch4.html

62 Smith, H. A., McKeen, J. D., & Singh, S. (2006). Making knowledge work: five principles for action-oriented knowledge management. Knowledge Management Research & Practice, 4(2), 116-124.

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Database: Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations and Ovid MEDLINE(R) <1946 to Present>
Search Strategy:
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1 Return to Work/ (929)

2 Employment, Supported/ (965)

3 Work/ (18878)

4 Employment/ (39799)

5 Unemployment/ (5713)

6 Occupations/ (21163)

7 Workplace/ (15989)

8 Occupational Medicine/ (23036)

9 Occupational Health/ (28115)

10 Occupational Health Services/ (10004)

11 Sick Leave/ (4524)

12 Absenteeism/ (7929)

13 (return* adj3 (work* or dut*)).tw,kw. (9843)

14 (employable or employee* or employment or unemployment or unemployed).tw,kw. (85298)

15 (sick leave* or sickness* or absence* or absenteeism).tw,kw. (538140)

16 ("disability management" or "disability prevention").tw,kw. (450)

17 (occupation or occupations or vocation or vocations).tw,kw. (31579)

18 (employability or employable or employee*).tw,kw. (36230)

19 ("work* adj2 abilit*" or ('work* adj2 capacity*') or ('work* adj2 activit*') or ('work adj2 disability*') or ('work* adj2 rehab*') or ('work* adj2 status*') or ('work* adj2 retention*') or (work* adj2 sustain*) or (work adj2 maintain*) or workabilit*).tw,kw. (22653)

20 or/1-19 (781782)

21 Disability Evaluation/ (39236)

22 exp Self-Help Devices/ (9342)

23 "Equipment and Supplies"/ (22368)

24 (case manager or case management).tw,kw. (9233)

25 (disab* adj3 guidance).tw,kw. (23)

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26 (assistive adj3 (device* or technolog* or self-help)).tw,kw. (2812)

27 (self help adj3 device*).tw,kw. (64)

28 Work Capacity Evaluation/ (5423)

29 Vocational Guidance/ (2040)

30 (vocational adj3 (training* or placement* or counseling)).tw,kw. (1818)

31 ((work* or job? or jobsite* or occupation* or employ*) adj5 interven*).tw,kw. (11855)

32 ((work* or job? or jobsite* or occupation* or employ*) adj5 accomod*).tw,kw. (22)

33 ((work* or job? or jobsite* or occupation* or employ*) adj5 modif*).tw,kw. (7043)

34 ((work* or job? or jobsite* or occupation* or employ*) adj5 adjust*).tw,kw. (5420)

35 ((work* or job? or jobsite* or occupation* or employ*) adj5 visit*).tw,kw. (3035)

36 ((work* or job? or jobsite* or occupation* or employ*) adj5 evaluat*).tw,kw. (32035)

37 ((work* or job? or jobsite* or occupation* or employ*) adj5 access*).tw,kw. (4463)

38 (reduced adj3 (employ* or occupation* or work* or duty or duties)).tw,kw. (3484)

39 (vocational adj3 guidance).tw,kw. (195)

40 (ergonomic adj3 (intervention? or counsel* or education or training or approach*)).tw,kw. (606)

41 ((light or modified or graded or alternative) adj3 (duty or duties or work*)).tw,kw. (4152)

42 or/21-41 (154976)

43 exp brain injuries/ (55732)

44 Craniocerebral Trauma/ (20483)

45 exp Head Injuries, Closed/ (8476)

46 exp Skull Fractures/ (19769)

47 mTBI*2.tw,kw. (1415)

48 tbi*2.tw,kw. (18022)

49 wrTBI*2.tw,kw. (8)

50 concuss*.tw,kw. (5252)

51 ((head* or cerebr* or crani* or capitis* or brain* or forebrain* or skull* or hemispher* or intracran* or orbit*) adj2 (injur* or trauma* or lesion* or damag* or wound* or destruction* or swell* or oedema* or edema* or fracture* or contusion* or commotion* or pressur*)).tw,kw. (158249)

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52 ((brain* or cerebr* or intracerebr* or crani* or intracran* or head* or subarachnoid* or subdural* or epidural* or extradural*) adj2 (haematoma* or hematoma* or hemorrhag* or haemorrhag* or pressur* or bleed*)).tw,kw. (76249)

53 or/43-52 (247310)

54 20 and 42 and 53 (440)

55 limit 54 to english language (369)

56 exp animals/ not (exp animals/ and exp humans/) (4244639)

57 55 not 56 (366)

58 (exp child/ or exp infant/) not ((exp child/ or exp infant/) and exp adult/) (1462717)

59 57 not 58 (361)

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Table 2| PRISMA-P (preferred reporting items for systematic review and meta-analysis protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Page No	Item No	Checklist item
Administrative information			
Title:			
Identification	1	1a	Identify the report as a protocol of a systematic review
Update	N/A	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	5	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:			
Contact	1	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	10	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	N/A	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:			
Sources	10	5a	Indicate sources of financial or other support for the review
Sponsor	10	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	10	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
Introduction:			
Rationale	3&4	6	Describe the rationale for the review in the context of what is already known
Objective	4	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
Methods:			
Eligibility criteria	5&6	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	5&6	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	Attached	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated

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Table 2| PRISMA-P (preferred reporting items for systematic review and meta-analysis protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Page No	Item No	Checklist item
Study records:			
Data management	6	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review
Selection process	5&6	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phaseof the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	6	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	6	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	6	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	6	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	7	15a	Describe criteria under which study data will be quantitatively synthesised
	7	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)
	7	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	7	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	6	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidenc	7	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

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Assessing the effectiveness of workplace accommodations in facilitating return-to-work after traumatic brain injury – a systematic review protocol.

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Assessing the effectiveness of workplace accommodations in facilitating return-to-work after traumatic brain injury – a systematic review protocol.

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Introduction: Returning to work and sustaining employment is a significant challenge for brain injury survivors. Within the literature, there is recurring support for the role of workplace accommodations in effective and early return-to-work (RTW). To date, however, there has been a lack of systematic reviews exploring the specific role of workplace accommodations within the context of RTW after brain injury. The primary objective of this protocol is to outline the methodological approach that will be undertaken to systematically review the literature and to assess the effectiveness of workplace accommodations in facilitating RTW.

Methods and Analysis: A total of nine databases will be searched systematically using the concepts 'Brain injury,' 'Return to work,' and 'Job Accommodations.' Study selection will be performed independently by three reviewers, based on predetermined eligibility criteria through two rounds of screening using, first, the title and abstract, followed by a full-text search. Extracted information will include the study's purpose, design, and setting; the data source and type; the severity of Traumatic Brain Injury (TBI) and the diagnostic criterion utilized; a comprehensive description of the intervention provided; the RTW outcome variables; and the statistical methods utilized etc. The data will be tabulated and narratively synthesized. Systematic review registration: This protocol has been registered with PROSPERO.

Ethics and Dissemination: As this review intends to use pre-existing published studies, ethical permissions will not be required. Nevertheless, this review will follow the ethical and governance standards in the data management and presentation of results. The findings from this review will potentially be published in a peer-reviewed scientific journal (electronically and in print). The results of this review will be presented at both national/international conferences and shared with stakeholders influencing return-to-work practices.

Keywords: Brain injury, Workplace accommodations, Return-to-work, Rehabilitation, Systematic review

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Strengths of this review:

- This focused review addresses the critical knowledge gap of how workplace accommodations can influence return-to-work outcomes and which category of accommodations have a greater impact on the sustenance of this return in the traumatic brain injury population has not been undertaken to date.
- Impact on health practitioners (physicians, rehabilitation professional, vocational counselors etc.) researchers, policy makers and survivors alike, i.e., acquisition of implementable knowledge regarding the nature of the accommodations and their impact; directing resources towards evidence-based and cost-effective solutions; engaging brain injury survivors to trial such accommodations early in their rehabilitation journey etc.
- Exploration using a sex/gender lens offers a distinct advantage to understand the sex- and gender-based influences on vocational outcomes.

Limitations:

- Heterogeneity of the traumatic brain injury population, uniqueness of the employer/occupational variables and individualized preferences and specific needs of the worker may pose a challenge for more precise interpretations.
- Environmental variables such as country of origin and, more specifically, variations in disability provisions, will need to be considered when interpreting the findings.The limited number of high level evidence within the vocational rehabilitation literature.

Implications:

Effective workplace accommodations promote an inclusive labor market by removing barriers. Utilizing the most effective workplace accommodations can provide employment opportunities to people with disabilities and vulnerabilities.

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BACKGROUND

Return-to-work (RTW) post-injury is a primary rehabilitation goal and perhaps the most reliable marker of functional recovery within a real-world context [1]. Returning to work and sustaining employment is a significant challenge for brain injury survivors [2]. It is estimated that 60% of working age individuals with moderate to severe acquired brain injury are unable to RTW post-injury [3] and a substantial proportion cannot sustain RTW. There is evidence that early and targeted vocational rehabilitation is cost-effective, and leads to improved work-related outcomes [2]. Since traumatic brain injury (TBI) more frequently occurs during prime working years, when people aim for vocational goals and are acquiring the skills needed for success [4], poor employment outcomes constitute a global health issue producing significant financial and societal burdens. The indirect economic cost of a TBI for work-aged Canadians is expected to increase to $8.2 billion by the year 2031, far exceeding costs of other common neurological conditions such as Spinal Cord Injury, Parkinson's Disease and Multiple Sclerosis [5].

TBI is caused by "a bump, blow, or jolt to the head or a penetrating head injury that disrupts the normal function of the brain" [6]. TBI is rated as mild, moderate and severe based on the initial trauma that caused the injury. Concussions and mild TBI (mTBI) are the most common of all TBIs [7], yet there is lack of understanding regarding their management in the workplace. mTBIs and concussions can have long-term functional effects on activities of daily living and return to productivity [8]. This is evidenced by the fact that the number of time loss claims for work-related concussions has increased by over 400% in Ontario, Canada, since 2005 [9]. The sequelea resulting from TBI are often diverse (i.e., headaches, memory difficulties, seizures, fatigue, visual and auditory deficits etc.) and accommodating for these impairments is necessary to ensure successful functioning in any re-integrative setting, including academic environments [10,11]. While memory loss and attention/concentration concerns are the most common functional limitations reported by TBI survivors, the persistent difficulties with emotional deregulation and the presence of medical symptoms have been identified as strong indicators of job retention [12].

Workplace accommodations (WAs) refer to the individually-tailored supports resulting from adaptations, alterations and/or modifications to job demands or workplace operations, which enable employee's equal access to essential aspects of their work [13, 14]. Within the context of personal injury, WAs attempt to facilitate re-integration to pre-injury work status [15]. WAs can be implemented in various forms across all stages of recovery to facilitate RTW. They may include provisions for flex days, opportunities for untimed breaks, the option to work from home, the arrangement of a job coach for specific work-oriented tasks etc [16,17].

Within the literature, there is recurring support for the role of WAs in effective and early RTW (i.e., shorter work absence duration [18], reduced disability costs [19], postponement or prevention of job loss, fewer job disruptions [20] and reduced productivity loss [21]). It is estimated that a significant proportion (83.5%) of workers with mental disorders need accommodations, yet only 30.5% receive them [22]. Workers who accept WAs are not only able to RTW but are also able to sustain work, compared to employees who rejected or did not receive an offer to accommodate [23]. In general, the use of compensatory strategies and supported employment practices enhance the likelihood of employment success [24]. Colantonio and colleagues [25] have shown that 76% of brain injury survivors have reported 'job modifications' and 'employers who accommodated treatment needs' as significant factors that facilitated RTW. More recently, Stergiou-Kita and colleagues have indicated that there is a lack of adequate understanding regarding the scope of WAs amongst both stakeholders, employers

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and employees [26,27]. A systematic review [28] did not find strong evidence to support that the patients' pre- or post-injury characteristics, or rehabilitation interventions has any effect on productivity status and vocational outcomes after TBI. Yet, the review did not consider organizational or vocational modifications such as accommodations and natural supports. Kristman and Colleagues [29], however, later reported that organizational factors are more important than demographic and occupational factors when considering RTW among employees with injuries other than TBI. Non-traditional determinants, such as family support, have been shown to play an equally important role in facilitating community and work re-integration [30].

There is evidence to suggest that sex/gender exerts significant influences on RTW processes, yet the influence of sex/gender on workplace injuries and outcomes is understudied [31,32]. Recent evidence reveals low and inconsistent support for the argument that gender significantly informs RTW after TBI [33, 34, 35], nevertheless positive outcome related to females in resuming work status after severe TBI is present [36]. After adjusting for demographic and health characteristics, adverse life events independently predicted increased rates of sickness-related absences among women [37]. It has been shown that while pink-collar employees are significantly more likely to receive an offer for accommodation, they are also more likely to refuse accommodations compared to their blue-collared counterparts [38]. The reasons for this discrepancy remain to be explored. A systematic review [39] on prognostic indictors of mTBI outcomes found that only 7% of >200 studies reviewed provided data stratified by sex. We are not aware of any other recent systematic reviews on RTW after TBI wherein the evidence was stratified by sex/gender.

To date, there is a lack of systematic reviews exploring the specific role of WAs within the context of RTW after TBI. While there are ample reviews that have investigated the role of work-related interventions on RTW, these reviews have focused on care and therapeutic programs or multidisciplinary intervention(s) that impact RTW including vocational rehabilitation. A more recent systematic review [40] found strong evidence that interventions with a combination of work-directed components are facilitating RTW in brain injury survivors. Nevertheless, the extracted data in this particular review paired rehabilitation interventions with work-related supports/aids to explain combined effects, versus stratifying for specific outcomes. Another scoping review concluded that WAs appear to be important for supporting employees with mental illness, but identified the implementation and evaluation of WAs as an area for further research [41]. Hence, while it is important to understand WAs from policy, legislative and social perspectives, it is perhaps more important to investigate the direct impact of WAs on employment re-entry and retention. Also, it is important to investigate the influence of sex/gender on the offer, acceptance and implementation of WAs.

The primary purpose of this protocol is to outline the methodology for a project that will systematically review the literature to identify the types of WAs that best facilitates re-entry to work and investigate the translational effect of successful WAs on the effectiveness of vocational outcomes, specifically RTW and, more importantly, the long-term sustenance of this return. The secondary aim is to consolidate and synthesize the quantitative and qualitative evidence in the literature separately. This review's final goal is to determine whether sex/gender variables are associated with the effectiveness of accommodations in promoting re-entry into the workplace.

METHODS/DESIGN

PROSPERO

This protocol has been registered with PROSPERO (PROSPERO 2016:CRD42016043517)

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PATIENT AND PUBLIC INVOLVEMENT

No patient involved.

SEARCH STRATEGY

The search strategy was developed with an information specialist at the Toronto Rehabilitation Institute-University Health Network (TRI-UHN). A modified PICO framework was used to facilitate and refine the search strategy. The search strategy was developed using a combination of comprehensive and specific subject headings and keywords, to balance sensitivity with specificity, and render articles across a wide spectrum of specific literature. Broader terms such as 'Return to Work', "Employment" and "Workplace" were used to search for articles, while specific narrow-focussed keywords such as "absenteeism", and "work" with "ability" and "capacity" were used to search relevant non-traditional publications. Similarly, generic heading such as "equipment and supplies" and "job accommodations" were searched alongside alternatives including "jobsite" with "interventions" and "duties." These alternative terms helped capture publications that may exclude more commonly deployed workplace adaptation keywords. Search terms were also adapted from previously published systematic reviews on the topic [42,43].

The search terms and strategy were independently peer-reviewed by another information specialist. Changes and suggestion were incorporated before the search terms were translated across nine databases, i.e., 1) Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations, 2) EBM Reviews Cochrane Central Register of Controlled Trials, 3) EBM Reviews Cochrane Database of Systematic Reviews, 4) Embase, 5) PsycINFO, 6) CINAHL, 7) Pubmed, 8) OT SEEKER, and 9) Health and Safety Science Abstracts. Additional supplementary file 1 provides the search strategy for each database. Furthermore, conference materials such as abstracts, editorials and/or commentaries will be excluded as they do not provide sufficient information required for the purposes of this review. All database results will be limited to English language. A limit to human only studies will be applied.

The grey literature will be searched for additional information using both 'Grey Matters' (a practical search tool for evidence based medicine) and resources from known relevant rehabilitation research groups (i.e., Institute of Work and Health, Job Accommodations Network, Employment Accessibility Resource Network etc). The reference lists of all included articles will be hand searched.

STUDY SELECTION

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for systematic reviews will be used to report the findings of this review [44].

Study selection will be completed by three reviewers, pre-determined eligibility criteria will guide all three reviewers and screening will be undertaken in two steps. Level I screen will involve screening of studies based on their titles and abstracts only. The Level I screen inclusion criteria are: 1) study participants must have sustained a TBI (irrespective of the mechanism or place of injury), 2) an outcome related to RTW must have been reported, and 3) study participants must be of working age (18-65 years). The primary outcome of interest, 'return-to-work', is defined as a post-injury attempt to resume productive activities. Each article may have one of three possible rating, i.e., 'yes', 'no' and 'maybe'. All three reviewers will discuss articles that have non-matched ratings; if consensus is not achieved, mediation consultation will be

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undertaken. Articles that pass the first screen criteria will proceed to the full text Level II screen. The Level II screen entails a full-text screening of articles. Two additional criteria will be considered during the full-text Level II screen: 1) study participants were working at the time of the injury/loss, and 2) an intervention that includes WAs (of any kind) was provided to facilitate the RTW.

For the purposes of this review, our operational definition of 'accommodation' is "any measurable change and/or alteration in the process flow of an organization, material used to accomplish a job demand, services provided/utilized at the workplace and/or routines for oneself (or others) to compensate for impairments or modify a work environment after a condition, disorder, injury and/or illness in an attempt to facilitate re-entry into the workforce" [Adapted from [45, 46]]. All study designs will be included, for instance, randomized and non-randomized controlled trials, case-control studies, cohort and longitudinal studies, retrospective analysis, scoping or systematic reviews, case studies, qualitative studies and/or mixed methods research studies. Qualitative and case studies will particularly be included as they may contain details which quantitative exploration alone cannot capture. With respect to vocational status, all types of employment (full-time, part-time, casual, contract, self-employed, students, volunteer etc.) and occupational categories will be considered. The same will be true for injury severity, whereby all mild, moderate and severe injuries will be considered. With sufficient data, analysis will be stratified by sex, age and injury severity. Data from international publications, albeit written in English, will be included.

DATA EXTRACTION AND MANAGEMENT

All three reviewers will develop and pilot the data extraction form. The form will be modified to ensure that relevant data is captured for the assessment of study quality. The data extraction form will include author(s), study title, publication year, purpose of study, study design, study setting, sample size, demographic and vocational characteristics of the study participants, data source or type (e.g., retrospective, interviews etc), severity of TBI and the diagnostic criterion utilized, statistical methodology utilized (e.g., descriptive, analytic and/or stratification techniques), comprehensive description of the intervention provided (e.g., employer-initiated, RTW co-ordinator, occupational therapist, case manager etc), duration of RTW, subsequent follow-up(s), characteristics and costs of the accommodations provided, outcome of the stated intervention etc. If possible, secondary outcomes such as quality of life, health status and satisfaction indicators will also be reported in a separate table. The stated conflicts of interest and 'Risk of Bias' assessments will also be presented in a table.

Study data included after the level II screen will be extracted by two reviewers using a blinded extraction process. Reviewers will cross-review the collected data to document and address any discrepancies and rationales. Following this comparative analysis, the revised data extraction will be peer-reviewed by the third reviewer. Completed data extraction forms will be archived, and information from these forms will be inputted and stored in Excel. One reviewer will independently check the data entry for accuracy.

QUALITY ASSESSMENT

The Critical Appraisal Skills Programme (CASP) checklists [47] will be used to assess study quality by guiding the assessment of validity and reliability of the included studies. The rationale for selecting the CASP checklists was made primarily due to the availability of separate checklists for different study designs such as cohort studies, review papers, randomized

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controlled trials and qualitative studies. Two reviewers will complete the checklists, through a blinded process, and later cross-review to document and address any discrepancies and rationales. Following consensus, the third reviewer will peer-review all checklists associated with this review's articles. With respect to missing data, study authors may be contacted for any queries regarding relevant data capture or additional data required to assess the risk of bias.

DATA ANALYSIS, SYNTHESIS AND REPORTING

Data will be collected from each study and tabulated in a summary table. Due to the anticipated heterogeneity of study designs, patient cohorts and variability of outcome measure, a meta-analytical approach may not be feasible. Nevertheless, descriptive statistics will be utilized to synthesize the evidence from quantitative studies while data from qualitative studies will be summarized by identifying prominent and prevalent themes. This approach is intended to enhance the utility and impact of the findings [48]. A data analysis strategy [49] may be used to aggregate results, integrate the original meaning of the primary studies and reconstruct the findings to facilitate new interpretations of the data. The sources of the synthesized information will be reported separately, however, attempts to mix and combine the results of both quantitative and qualitative data will be made to provide a comprehensive conceptual understanding of the findings.

Sub-group analysis: With sufficient data, sub-group analyses will be undertaken to investigate differences in outcomes. Given the expansion of the literature with respect to sex/gender differences in outcomes, two relevant Cochrane tools will be utilized, i.e., the Sex and Gender in Systematic Reviews Planning Tool and the Equity Checklist for Systematic Review Authors [50,51]. This unique investigation will clarify differences observed in data collection, analysis, and interpretation across studies. To examine the impact of interventions on inequality, the data will be further assessed using an intersectional lens to better understand the contributions of social identities (e.g., sex, age, race, socioeconomic status, etc.) as well as disadvantaged populations on the outcomes of RTW and productivity.

The review will be undertaken according to this published protocol. Any deviations from the protocol will be reported in a separate sub-section and labeled "Differences between published protocol and review."

DISCUSSION

We believe this focused undertaking is the first of its kind. At this time, there have been no systematic reviews exploring the effectiveness of specific WAs that facilitate and/or sustain RTW after TBI. Reviews published to date have considered the role of a combination of care programs such as outpatient rehabilitation programs, generic workplace interventions, the impact of community-based remediation on RTW and/or return-to-productivity outcomes.

The Convention on the Rights of Persons with Disabilities (CRPD) [52], Disability Discrimination Act (DDA) [53] as well as the Canadian Human Rights Act (CHRC) [14] mandates equal rights and opportunities at the workplace, i.e., employees with disabilities are entitled to the same opportunities and benefits as people without disabilities. Under the legal test, CHRC, the employer has a duty to take reasonable steps to accommodate the position of an employee unless the employer's business would suffer 'undue hardship'. Although the Guides for Assessing Persons with Disabilities [45] have provided some guidance regarding the interpretation of undue hardship, operationalizing the construct of WAs has been short of easy. We believe this review will help shed light on the various conceptualizations of accommodations within the workplace

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and how WAs are operationalized within the practice of RTW following TBI. Investigating effective interventions that facilitate returning TBI survivors to work not only has the potential for short-term impact on a person's working life and ability to contribute to the economy, but also on the long-term health and well-being of society.

Given the evolving workforce, certain work-related supports are naturally embedded within the workplace, available for all employees and not considered a customized adjustment. These are often referred to as 'natural supports'. It is important to distinguish between WAs and natural supports, the former being a support customized for a particular worker who is encountering difficulties with work resumption. This review, however, will be focused specifically on WAs.

Effective WAs holds the potential to foster inclusive work designs for TBI survivors who are considering a return to employment. Given that brain injury occurs frequently among working age individuals, the study has implications from a labor market policy perspective. The information gathered and consolidated in this review will help health practitioners, policy makers, researchers and other workforce-related stakeholders make informed decisions when considering service delivery or future research into RTW for brain injury survivors. With respect to service delivery, the 'low cost and high impact'[54] feature of WAs makes it not only easy to implement but also economically lucritive. Added benefits of WAs include retention of previously trained employees and hence elimination of costs associated with training a new employee, boosted morale amongst workers and an increased sense of well-being at both individual and societal levels.

Traditional RTW practices focus excessively on sickness and disability, whereby, based on a reported mismatch between the client's limitations and the demands of the job - a worker is deemed "unfit to work". In 2010, the United Kingdom replaced its "sickness certificate" with a statement of fitness for work also known as the "fit note" [55]. The shift in methodology from "not fit to work" to "fit to work" (if the employer is able to make necessary modifications) transformed the traditional "disability" lens into a "functional" focus. Currently, many procedural, legislative, training and practical issues remain regarding the nature of the ideal fit note, nevertheless, there is evidence to support that these notes, when filled correctly, are not only useful to the employer for planning an employee's return but are also improving RTW rates by more than 80% [56].In countries where reasonable workplace accommodations are legislated, health Care Practitioner's, Physician's and employer's knowledge of effective WAs can facilitate work re-entry by ensuring an inclusive and safe work environment for the worker.

Also, the findings of this review are transferable across other sectors concerned with RTW or sustainability at work. We will provide a few examples for clarity's sake. First, the review aims to investigate the reported sex/gender based differences within the context of WAs and RTW. Since women, on average, have shorter earning careers and interruptions related to motherhood and unequal sharing of care/housework demands and early retirement [57,58], effective WAs will help them sustain employment longevity to support retirement/pension planning. Vocational longevity also helps facilitates continuous employment through various stages of a female's career (i.e., childbearing, life transitions) and combats the female wage inequality gap. Second, workplaces have to become more accommodating of older workers particularly owing to age-related cognitive decline. By 2031, approximately 25% of Canada's population is expected to be age 65 or older [59]. Moreover, Canadians are working longer and retiring later. Over the past few decades, the average retirement age has risen from 61 to 63 years

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[60]. Similar trends are observed in the United Kingdom [61]. Effective WAs may lessen the economic burden associated with TBI.

Effective WAs, for workers with special needs, may foster increased productivity which leads to economic well-being and eventually can be operationalized as the living standards for the population. Creating an inclusive labor market by removing barriers and providing employment opportunities to people with disabilities will not only generate a high impact return on the investment but also maximize the utility of human capital.

DISSEMINATION PLAN

In addition to traditional knowledge translation approaches (i.e., publishing the findings in peer-reviewed journals, presenting at local and international conferences, posting summary-of-findings on university websites etc), findings from this review will be disseminated using an integrated action-oriented knowledge translation framework [62]. It is hoped that this action-oriented framework will enable contextualization of the knowledge and findings to ensure impact in the real-world. Initial interest will be ignited through free webinars and podcasts as well as social media engagements, such as Twitter. Later, various stakeholders (i.e., Workplace Safety and Insurance Board, Insurance Bureau of Canada, Accessibilty Ontario etc.) will be invited to engage in a discussion to: 1) understand and contextualize the knowledge gained and 2) to explore how to reduce potential uptake barriers within the workplace environment. A special rehabilitation focus will be maintained to translate the knowledge to health practitioners, and to facilitate the development guidelines that will help facilitate the integration of these findings into day-to-day RTW practice.

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Contributors: AC and RS conceptualized the study. RS formulated the design, drafted the manuscript, assisted with the development of the search terms and registered the protocol via PROSPERO. AC provided expertise at each level and also reviewed the protocol. Both authors read and approved the final manuscript.

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References

1 Franulic, A., Carbonell, C. G., Pinto, P., & Sepulveda, I. (2004) Psychosocial adjustment and employment outcome 2, 5 and 10 years after TBI, Brain Injury, 18:2, 119-129

2 Radford, K., Phillips, J., Drummond, A., Sach, T., Walker, M., Tyerman, A., Haboubi, N., & Jones, T. (2013). Return to work after traumatic brain injury: cohort comparison and economic evaluation. Brain Inj, 27(5), 507-520.

3 van Velzen, J. M., van Bennekom, C. A., Edelaar, M. J., Sluiter, J. K., & Frings-Dresen, M. H. (2009). How many people return to work after acquired brain injury?: A systematic review, Brain Injury, 23:6, 473-488

4 Sarajuuri, J. M., Kaipio, M. L., Koskinen, S. K., Niemela, M. R., Servo, A. R., & Vilkki, J. S. (2005). Outcome of a comprehensive neurorehabilitation program for patients with traumatic brain injury. Arch Phys Med Rehabil, 86(12), 2296-2302.

5 Public Health Agency of Canada and Health Canada Canada. (2014). Mapping Connections: An Understanding of Neurological Conditions in Canada. Retrieved from: http://www.phac-aspc.gc.ca/publicat/cd-mc/mc-ec/index-eng.php

6 Centers for Disease Control and Prevention (2016). TBI: Get the Facts. Retrieved from: http://www.cdc.gov/traumaticbraininjury/get_the_facts.html

7 National Center for Injury Prevention and Control. Report to Congress on Mild Traumatic Brain Injury in the United States: Steps to Prevent a Serious Public Health Problem. Atlanta, GA: Centers for Disease Control and Prevention; 2003.

8 Cancelliere, C., Kristman, V. L., Cassidy, D., Hincapié, C. A., Côté, P., Boyle, E., Carroll, L. J., Stålnacke, B., Boussard, c. N., & Borg, J. (2014). Systematic Review of Return to Work After Mild Traumatic Brain Injury: Results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Archives of Physical Medicine and Rehabilitation 95(3): S201-S209.

9 Workplace Safety & Insurance Board (2015). By The Numbers: 2014 WSIB Statistical Report. Retrieved from: http://www.wsibstatistics.ca/

10 Jantz, P. B., & Coulter, G.A. (2007). Child and adolescent traumatic brain injury: Academic, behavioural, and social consequences in the classroom. Support for Learning, 22(2), 84-89.

11 Hendricks, D. J., Sampson, E., Rumrill, P., Leopold, A., Elias, E., Jacobs, K., Nardoned, A., Scherere, M., & Stauffer, C. (2015). Activities and interim outcomes of a multi-site development project to promote cognitive support technology use and employment success among postsecondary students with traumatic brain injuries. NeuroRehabilitation, 37(3), 449-458.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12 Artman L. K., & McMahon B.T. (2013). Functional limitations in TBI and their relationship to job maintenance following work re-entry. Journal of Vocational Rehabilitation 39, 39, 13–21.

13 Americans With Disabilities Act [Pub. L. No. 101-336, 104 Stat. 328, 1990].

14 Canadian Human Rights Act [R.S.C., 1985, c. H-6].

15 Mueller, J. L. (1999). Returning to work through job accommodation: a case study. AAOHN Journal, 47(3), 120-129; quiz 120-131.

16 Targett, P., & Wehman, P. (2010) Return to work after traumatic brain injury: a supported employment approach. In: Schultz IZ, Rogers ES (eds) Handbook of work accommodatins and retention in mental health. Springer, New York

17 Schultz, I. Z., Winter, A., & Wald, J. (2010) Evidentiary support for best practices in job accommodation in mental health: macrosystem, employer-level and employee-level interventions. In: Schultz, I. Z., Rogers, E. S. (eds) Handbook of work accommodation and retention in mental health. Springer, New York

18 Franche, R. L., Cullen, K., Clarke, J., Irvin, E., Sinclair, S., Frank, J., & Institute for Work & Health (IWH) Workplace-Based RTW Intervention Literature Review Research Team. (2005). Workplace-based return-to-work interventions: a systematic review of the quantitative literature. J Occup Rehabil, 15(4), 607-631.

19 Franche, R. L., Severin, C. N., Hogg-Johnson, S., Cote, P., Vidmar, M., & Lee, H. (2007). The impact of early workplace-based return-to-work strategies on work absence duration: a 6-month longitudinal study following an occupational musculoskeletal injury. J Occup Environ Med, 49(9), 960-974.

20 Johnson, K. L., Yorkston, K. M., Klasner, E. R., Kuehn, C. M., Johnson, E., & Amtmann, D. (2004). The cost and benefits of employment: a qualitative study of experiences of persons with multiple sclerosis. Arch Phys Med Rehabil, 85(2), 201-209.

21 Gignac, M. A., Cao, X., & McAlpine, J. (2015). Availability, need for, and use of work accommodations and benefits: are they related to employment outcomes in people with arthritis? Arthritis Care Res (Hoboken), 67(6), 855-864.

22 Wang, J., Patten, S., Currie, S., Sareen, J., & Schmitz, N. (2011). Perceived needs for and use of workplace accommodations by individuals with a depressive and/or anxiety disorder. J Occup Environ Med, 53(11), 1268-1272.

23 Franche, R. L., Severin, C. N., Hogg-Johnson, S., Lee, H., Cote, P., & Krause, N. (2009). A multivariate analysis of factors associated with early offer and acceptance of a work

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

accommodation following an occupational musculoskeletal injury. J Occup Environ Med, 51(8), 969-983.

24 Target, P., Wehman, P., Petersen, R., & Gorton, S. (1998). Enhancing work outcome for three persons with traumatic brain injury. Int J Rehabil Res, 21(1), 41-50.

25 Colantonio, A., & Lee, V. (2012). Return to work after mild to moderate work related injury: A pilot study. Brain Injury, 26 (4-5), 778.

26 Stergiou-Kita, M., Mansfield, E., & Colantonio, A. (2014). Injured workers' perspectives on how workplace accommodations are conceptualized and delivered following electrical injuries. J Occup Rehabil, 24(2), 173-188.

27 Stergiou-Kita, M., Daiter, L., & Colantonio, A. (2015). Good Intentions? Employer Representative Conceptualizations and, Challenges to the Workplace Accommodation Process: The Case of Electrical Injuries. Employ Respons Rights J, 27, 1–25.

28 Saltychev, M., Eskola, M., Tenovuo, O., & Laimi, K. (2013). Return to work after traumatic brain injury: Systematic review. Brain Inj, 27(13-14), 1516-1527.

29 Kristman, V. L., Shaw, W. S., Reguly, P., Williams-Whitt, K., Soklaridis, S., & Loisel, P. (2016). Supervisor and Organizational Factors Associated with Supervisor Support of Job Accommodations for Low Back Injured Workers. J Occup Rehabil.

30 Gagnon, A., Lin, J., & Stergiou-Kita, M. (2016). Family members facilitating community re-integration and return to productivity following traumatic brain injury - motivations, roles and challenges. Disabil Rehabil, 38(5), 433-441.

31 Stergiou-Kita, M., Mansfield, E., Sokoloff, S., & Colantonio, A. (2016). Gender Influences on Return to Work After Mild Traumatic Brain Injury. Archives of Physical Medicine and Rehabilitation, 97(2 Suppl), S40-45.

32 Colantonio, A. (2016). Sex, Gender, and Traumatic Brain Injury: A Commentary. Archives of Physical Medicine and Rehabilitation, Volume 97 , Issue 2 , S1 - S4

33 Keyser-Marcus, L. A., Bricout, J. C., Wehman, P., Campbell, L. R., Cifu, D. X., Englander, J., High, W., & Zafonte, R. D. (2002). Acute predictors of return to employment after traumatic brain injury: A longitudinal follow-up. Archives of Physical Medicine and Rehabilitation, 83(5), 635-641.

34 Kreutzer, J. S., Marwitz, J. H., Walker, W., Sander, A., Sherer, M., Bogner, J., Fraser, R., & Bushnik, T. (2003). Moderating factors in return to work and job stability after traumatic brain injury. J Head Trauma Rehabil, 18(2), 128-138.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

35 Ownsworth, T., & McKenna, K. (2004). Investigation of factors related to employment outcome following traumatic brain injury: a critical review and conceptual model. Disabil Rehabil, 26(13), 765-783.

36 Scherzer, F. C. P. (1993). Predictors and indicators of work status after traumatic brain injury: A meta-analysis. . Neuropsychological Rehabilitation, 3(1), 5-35.

37 Suominen, S., Vahtera, J., Korkeila, K., Helenius, H., Kivimaki, M., & Koskenvuo, M. (2007). Job strain, life events, and sickness absence: a longitudinal cohort study in a random population sample. J Occup Environ Med, 49(9), 990-996.

38 Franche, R. L., Severin, C. N., Hogg-Johnson, S., Lee, H., Cote, P., & Krause, N. (2009). A multivariate analysis of factors associated with early offer and acceptance of a work accommodation following an occupational musculoskeletal injury. J Occup Environ Med, 51(8), 969-983.

39 Cancelliere, C., Donovan, J., & Cassidy, J. D. (2016). Is Sex an Indicator of Prognosis After Mild Traumatic Brain Injury: A Systematic Analysis of the Findings of the World Health Organization Collaborating Centre Task Force on Mild Traumatic Brain Injury and the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch Phys Med Rehabil, 97(2 Suppl), S5-18.

40 Donker-Cools, B. H. P. M., Daams, J. G., Wind, H., & Frings-Dresen, M. H. W. (2016). Effective return-to-work interventions after acquired brain injury: A systematic review. Brain Injury, 30(2), 113-131.

41 McDowell, C., & Fossey, E. (2015). Workplace accommodations for people with mental illness: a scoping review. J Occup Rehabil, 25(1), 197-206.

42 Nieuwenhuijsen, K., Faber, B., Verbeek, J.H., Neumeyer-Gromen, A., Hees, H.L., Verhoeven, A.C., van der Feltz-Cornelis, C.M., and Bültmann, U.(2014). Interventions to improve return to work in depressed people. Cochrane Database Syst Rev, 3;(12): CD006237

43 van Vilsteren, M., van Oostrom, S. H., de Vet, H. C., Franche, R. L., Boot, C. R., and Anema, J.R. (2015). Workplace interventions to prevent work disability in workers on sick leave. Cochrane Database Syst Rev, 5(10):CD006955

44 Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., & Group, T.P. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.

45 Public Service Commission of Canada. (2007). Guide for Assessing Persons with Disabilities: How to determine and implement assessment accommodations. Retrieved from: http://www.psc-cfp.gc.ca/plcy-pltq/guides/assessment-evaluation/apwd-eph/pdf/apwd-eph-eng.pdf

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

46 Job Accommodations Network. Available at: http://askjan.org

47 Critical Appraisal Skills Programme (CASP) Checklists (2013). Retrieved from: http://www.casp-uk.net/casp-tools-checklists

48 Harden, A. (2010). Mixed-Methods Systematic Reviews: Integrating Quantitative and Qualitative Findings. Technical Brief no. 25.

49 Jang, E. E., McDougall, D. E., Herbert, M., Pollon, D., & Russell, P. (2008). Integrative mixed methods data analytic strategies in research on school success in challenging circumstances. Journal of Mixed Methods Research, 2, 221-247.

50 Doull M., Runnels V., Tudiver S., Boscoe M. (2011). Sex and gender in systematic reviews: Planning Tool.

51 Ueffing E, Tugwell P, Welch V, Petticrew M, Kristjansson E for the Campbell and Cochrane Equity Methods Group. Equity Checklist for Systematic Review Authors. Version 2012-10-02. Retrieved from: http://equity.cochrane.org/sites/equity.cochrane.org/files/uploads/EquityChecklist2012.pdf

52 United Nations General Assembly, Convention on the Rights of Persons with Disabilities [Resolution 61/106]. Retrieved from: http://www.ohchr.org

53 Disability Discrimination Act [1995, c. 50]. Retrieved from: http://www.legislation.gov.uk/ukpga/1995/50/contents

54 Office of Disability Employment Policy. (2007). Workplace Accommodations: Low Cost, High Impact. Retrieved from http://digitalcommons.ilr.cornell.edu/gladnetcollect/383

55 Department for Work and Pensions. (2013). Guidance for GPs, employers, hospital doctors and occupational health practitioners on using fit notes (statements of fitness for work).. Retrieved from: https://www.gov.uk/government/collections/fit-for-work-guidance

56 Coole, C., Potgieter, I., Nouri, F., Worthington, E., & Drummond, A. (2015). Return-to-work outcomes and usefulness of actual fit notes received by employers. Fam Pract, 32(5), 551-556.

57 Gower, D. (1997). "Measuring the age of retirement." Perspectives on Labour and Income: Statistics Canada, Catalogue no. 75-001-XPE, 9 (2)): 11-17.

58 Angelov, N., Johansson, P., and Lindahl, E. (2013). Is the Persistent Gender Gap in Income and Wages Due to Unequal Family Responsibilities?. IZA Discussion Paper No. 7181. Available at SSRN: https://ssrn.com/abstract=2210841

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59 Denton, F.T., Feaver, C.H. & Spencer, B.G. (2005). Time series analysis and stochastic forecasting: An econometric study of mortality and life expectancy. Journal of Population Economics18: 203-227

60 Statistics Canada. (2015). Study: The local unemployment rate and retirement, 1991 to 2007. Retrieved from: http://www.statcan.gc.ca/daily-quotidien/150422/dq150422a-eng.pdf

61 Office of National Statistics. (2013). Pension Trends – Chapter 4: The Labour Market and Retirement. Retrieved from: http://webarchive.nationalarchives.gov.uk/20160105160709/http://www.ons.gov.uk/ons/rel/pensions/pension-trends/chapter-4--the-labour-market-and-retirement--2013-edition/art-pt2013ch4.html

62 Smith, H. A., McKeen, J. D., & Singh, S. (2006). Making knowledge work: five principles for action-oriented knowledge management. Knowledge Management Research & Practice, 4(2), 116-124.

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Database: Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations and Ovid MEDLINE(R) <1946 to Present>
Search Strategy:
---

1 Return to Work/ (929)

2 Employment, Supported/ (965)

3 Work/ (18878)

4 Employment/ (39799)

5 Unemployment/ (5713)

6 Occupations/ (21163)

7 Workplace/ (15989)

8 Occupational Medicine/ (23036)

9 Occupational Health/ (28115)

10 Occupational Health Services/ (10004)

11 Sick Leave/ (4524)

12 Absenteeism/ (7929)

13 (return* adj3 (work* or dut*)).tw,kw. (9843)

14 (employable or employee* or employment or unemployment or unemployed).tw,kw. (85298)

15 (sick leave* or sickness* or absence* or absenteeism).tw,kw. (538140)

16 ("disability management" or "disability prevention").tw,kw. (450)

17 (occupation or occupations or vocation or vocations).tw,kw. (31579)

18 (employability or employable or employee*).tw,kw. (36230)

19 ("work* adj2 abilit*" or ('work* adj2 capacity*') or ('work* adj2 activit*') or ('work adj2 disability*') or ('work* adj2 rehab*') or ('work* adj2 status*') or ('work* adj2 retention*') or (work* adj2 sustain*) or (work adj2 maintain*) or workabilit*).tw,kw. (22653)

20 or/1-19 (781782)

21 Disability Evaluation/ (39236)

22 exp Self-Help Devices/ (9342)

23 "Equipment and Supplies"/ (22368)

24 (case manager or case management).tw,kw. (9233)

25 (disab* adj3 guidance).tw,kw. (23)

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26 (assistive adj3 (device* or technolog* or self-help)).tw,kw. (2812)

27 (self help adj3 device*).tw,kw. (64)

28 Work Capacity Evaluation/ (5423)

29 Vocational Guidance/ (2040)

30 (vocational adj3 (training* or placement* or counseling)).tw,kw. (1818)

31 ((work* or job? or jobsite* or occupation* or employ*) adj5 interven*).tw,kw. (11855)

32 ((work* or job? or jobsite* or occupation* or employ*) adj5 accomod*).tw,kw. (22)

33 ((work* or job? or jobsite* or occupation* or employ*) adj5 modif*).tw,kw. (7043)

34 ((work* or job? or jobsite* or occupation* or employ*) adj5 adjust*).tw,kw. (5420)

35 ((work* or job? or jobsite* or occupation* or employ*) adj5 visit*).tw,kw. (3035)

36 ((work* or job? or jobsite* or occupation* or employ*) adj5 evaluat*).tw,kw. (32035)

37 ((work* or job? or jobsite* or occupation* or employ*) adj5 access*).tw,kw. (4463)

38 (reduced adj3 (employ* or occupation* or work* or duty or duties)).tw,kw. (3484)

39 (vocational adj3 guidance).tw,kw. (195)

40 (ergonomic adj3 (intervention? or counsel* or education or training or approach*)).tw,kw. (606)

41 ((light or modified or graded or alternative) adj3 (duty or duties or work*)).tw,kw. (4152)

42 or/21-41 (154976)

43 exp brain injuries/ (55732)

44 Craniocerebral Trauma/ (20483)

45 exp Head Injuries, Closed/ (8476)

46 exp Skull Fractures/ (19769)

47 mTBI*2.tw,kw. (1415)

48 tbi*2.tw,kw. (18022)

49 wrTBI*2.tw,kw. (8)

50 concuss*.tw,kw. (5252)

51 ((head* or cerebr* or crani* or capitis* or brain* or forebrain* or skull* or hemispher* or intracran* or orbit*) adj2 (injur* or trauma* or lesion* or damag* or wound* or destruction* or swell* or oedema* or edema* or fracture* or contusion* or commotion* or pressur*)).tw,kw. (158249)

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52 ((brain* or cerebr* or intracerebr* or crani* or intracran* or head* or subarachnoid* or subdural* or epidural* or extradural*) adj2 (haematoma* or hematoma* or hemorrhag* or haemorrhag* or pressur* or bleed*)).tw,kw. (76249)

53 or/43-52 (247310)

54 20 and 42 and 53 (440)

55 limit 54 to english language (369)

56 exp animals/ not (exp animals/ and exp humans/) (4244639)

57 55 not 56 (366)

58 (exp child/ or exp infant/) not ((exp child/ or exp infant/) and exp adult/) (1462717)

59 57 not 58 (361)

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Table 2| PRISMA-P (preferred reporting items for systematic review and meta-analysis protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Page No	Item No	Checklist item
Administrative information			
Title:			
Identification	1	1a	Identify the report as a protocol of a systematic review
Update	N/A	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	5	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:			
Contact	1	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	10	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	N/A	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:			
Sources	10	5a	Indicate sources of financial or other support for the review
Sponsor	10	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	10	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
Introduction:			
Rationale	3&4	6	Describe the rationale for the review in the context of what is already known
Objective	4	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
Methods:			
Eligibility criteria	5&6	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	5&6	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	Attached	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated

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Table 2| PRISMA-P (preferred reporting items for systematic review and meta-analysis protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Page No	Item No	Checklist item
Study records:			
Data management	6	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review
Selection process	5&6	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phaseof the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	6	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	6	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	6	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	6	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	7	15a	Describe criteria under which study data will be quantitatively synthesised
	7	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)
	7	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	7	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	6	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidenc	7	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

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BMJ Open

Assessing the effectiveness of workplace accommodations in facilitating return-to-work after traumatic brain injury – a systematic review protocol.

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Assessing the effectiveness of workplace accommodations in facilitating return-to-work after traumatic brain injury – a systematic review protocol.

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Introduction: Returning to work and sustaining employment can be a significant challenge for traumatic brain injury (TBI) survivors. Within the literature, there is recurring support for the role of workplace accommodations in effective and early return-to-work (RTW). To date, however, there has been a lack of systematic reviews exploring the specific role of workplace accommodations within the context of RTW after TBI. The primary objective of this protocol is to outline the methodological approach that will be undertaken to systematically review the literature and to assess the effectiveness of workplace accommodations in facilitating RTW.

Methods and Analysis: A total of nine databases will be searched systematically using the concepts 'Brain injury,' 'Return to work,' and 'Job Accommodations.' Study selection will be performed independently by three reviewers, based on predetermined eligibility criteria through two rounds of screening using, first, the title and abstract, followed by a full-text search. Extracted information will include the study's purpose, design, and setting; the data source and type; the severity of TBI and the diagnostic criterion utilized; a comprehensive description of the intervention provided; the RTW outcome variables; and the statistical methods utilized etc. The data will be tabulated and narratively synthesized. Systematic review registration: This protocol has been registered with PROSPERO.

Ethics and Dissemination: As this review intends to use pre-existing published studies, ethical permissions will not be required. Nevertheless, this review will follow the ethical and governance standards in the data management and presentation of results. The findings from this review will potentially be published in a peer-reviewed scientific journal (electronically and in print). The results of this review will be presented at both national/international conferences and shared with stakeholders influencing return-to-work practices.

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Keywords: Traumatic Brain injury, Workplace accommodations, Return-to-work, Rehabilitation, Systematic review

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Strengths of this review:

- This focused review addresses critical knowledge gaps in understanding how workplace accommodations can influence return-to-work outcomes and which categories of workplace accommodations have a greater impact on the sustainability of RTW in the TBI population.
- Impact on health practitioners (physicians, rehabilitation professionals, vocational counselors etc.), researchers, policy makers and survivors alike, i.e., acquisition of implementable knowledge regarding the nature of the accommodations and their impact; directing resources towards evidence-based and cost-effective solutions; engaging TBI survivors to trial such accommodations early in their rehabilitation journey etc.
- Exploration using a sex and gender lens offers a distinct advantage to understand the sex- and gender-based influences on vocational outcomes.

Limitations:

- Heterogeneity of the TBI population, uniqueness of the employer/occupational variables as well as the individualized preferences and specific needs of the worker, as they relate to workplace accommodations, may pose a challenge for more precise interpretations.
- Environmental variables such as country of origin and, more specifically, variations in disability provisions, will need to be considered when interpreting the findings.

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BACKGROUND

Return-to-work (RTW) post-injury is a primary rehabilitation goal and perhaps the most reliable marker of functional recovery within a real-world context [1]. Returning to work and sustaining employment is a significant challenge for traumatic brain injury (TBI) survivors [2]. It is estimated that 60% of working age individuals with moderate to severe acquired brain injury are unable to RTW post-injury [3] and a substantial proportion cannot sustain RTW. There is evidence that early and targeted vocational rehabilitation is cost-effective, and leads to improved work-related outcomes [2]. Since TBI more frequently occurs during prime working years, when people aim for vocational goals and are acquiring the skills needed for success [4], poor employment outcomes constitute a global health issue producing significant financial and societal burdens. The indirect economic cost of a TBI for working age Canadians is expected to increase to $8.2 billion by the year 2031, far exceeding costs of other common neurological conditions such as spinal cord injury, Parkinson's disease and multiple sclerosis [5].

TBI is caused by "a bump, blow, or jolt to the head or a penetrating head injury that disrupts the normal function of the brain" [6]. TBI is rated as mild, moderate and severe based on the initial trauma that caused the injury. Concussion and mild TBI (mTBI) are the most common of all TBIs [7], yet there is lack of understanding regarding their management in the workplace. mTBIs and concussions can have long-term functional effects on activities of daily living and return to productivity [8]. This is evidenced by the fact that the number of time loss claims for work-related concussions has increased by over 400% in Ontario, Canada, from 2005 to 2013[9]. The sequelea resulting from TBI are often diverse (e.g., headaches, memory difficulties, seizures, fatigue, visual and auditory deficits etc.) and accommodating for these impairments is necessary to ensure successful functioning in any re-integrative setting, including academic environments [10 11]. While memory loss and attention/concentration concerns are the most common functional limitations reported by TBI survivors, the persistent difficulties with emotional deregulation and the presence of medical symptoms have been identified as strong indicators limiting job retention [12].

Workplace accommodations (WAs) refer to the individually-tailored supports resulting from adaptations, alterations and/or modifications to job demands or workplace operations, which enable employee's equal access to essential aspects of their work [13 14]. Within the context of personal injury, WAs attempt to facilitate re-integration to pre-injury work status [15]. WAs can be implemented in various forms across all stages of recovery to facilitate RTW. They may include provisions for flex days, opportunities for untimed breaks, the option to work from home, the arrangement of a job coach for specific work-oriented tasks etc [16 17].

Within the literature, there is recurring support for the role of WAs in effective and early RTW (e.g., shorter work absence duration [18], reduced disability costs [19], postponement or prevention of job loss, fewer job disruptions [20] and reduced productivity loss [21]). It is estimated that a significant proportion (83.5%) of workers with mental health conditions need accommodations, yet only 30.5% receive them [22]. After a musculoskeletal injury, workers who accept WAs are not only able to RTW but are also able to sustain work, compared to employees who rejected or did not receive an offer to accommodate [23]. In general, the use of compensatory strategies and supported employment practices enhance the likelihood of employment success after TBI [24]. Colantonio and colleagues [25] have reported that 76% of TBI survivors identified 'job modifications' and 'accommodations for treatment needs by employers' as factors that facilitated RTW. Stergiou-Kita and colleagues have indicated that there is a lack of adequate understanding regarding the scope of WAs amongst both stakeholders, employers

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and employees [26 27]. A systematic review [28] did not find strong evidence to support that the patients' pre- or post-injury characteristics, or rehabilitation interventions had any effect on productivity status and vocational outcomes after TBI. Yet, the review did not consider organizational or vocational modifications such as accommodations and natural supports. Kristman and Colleagues [29], however, later reported that organizational factors are more important than demographic and occupational factors when considering RTW among employees with injuries other than TBI. Non-traditional determinants, such as family support, have been shown to play an equally important role in facilitating community and work re-integration following TBI [30].

There is evidence to suggest that sex and gender exerts significant influences on RTW processes, yet the influence of sex and gender on workplace injuries and outcomes is understudied [31 32]. Further, these terms are often used interchangeably in the literature but they refer to different constructs. Recent evidence reveals low and inconsistent support for the argument that sex and gender significantly informs RTW after TBI [33-35] partly given the small number of females included in these samples; nevertheless positive outcome related to females in resuming work status after severe TBI is present [36]. After adjusting for demographic and health characteristics, adverse life events independently predicted increased rates of sickness-related absences among women [37] in a sex-balanced sample. Franche and colleagues have reported that, following an occupational musculoskeletal injury, pink-collar employees (referred to as female-dominated occupations such as clerical, sales and service) are significantly more likely to receive an offer for accommodation, they are also more likely to refuse accommodations compared to their blue-collared counterparts (referred to as male-dominated occupations such as manufacturing, materials handling, construction, transport, mining) [23]. The reasons for these discrepancies remain to be explored. A systematic review [38] on prognostic indicators of mTBI outcomes found that only 7% of >200 studies reviewed provided data stratified by sex. We are not aware of any other recent systematic reviews on RTW after TBI wherein the evidence was stratified by sex and gender.

To date, there has been a knowledge gap created by the lack of systematic reviews exploring the specific role of WAs within the context of RTW after TBI. While there are studies and reviews that have investigated the role of work-related interventions on RTW, these reviews have focused on care and therapeutic programs or multidisciplinary intervention(s) that impact RTW including vocational rehabilitation [39-41]. Donker-Cools and colleagues conducted a systematic review [42] and found strong evidence that interventions with a combination of work-directed components are facilitating RTW in TBI survivors. Nevertheless, the extracted data in this particular review paired rehabilitation interventions with work-related supports/aids to explain combined effects, versus stratifying for specific outcomes. Another scoping review concluded that WAs appear to be important for supporting employees with mental illness, but identified the implementation and evaluation of WAs as an area for further research [43]. A more recent systematic review focused on workplace factors, conducted by Alves and colleagues [44], identifies that onsite workplace adaptions are potential avenues for increased RTW yet there remains lack of a robust evidence base concerning the association between modifiable workplace factors and work retention after experiencing ABI; the scope of their work, however, was broad and included both traumatic and non-traumatic brain injuries. Hence, while it is important to understand WAs from policy, legislative and social perspectives, it is perhaps more paramount to investigate the direct impact of WAs on employment re-entry and retention. Also, it is important

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to investigate the influence of sex and gender on the offer, acceptance and implementation of WAs.

The primary purpose of this protocol is to outline the methodology for a project that will systematically review the literature to identify and characterize the categories of WAs that best facilitate re-entry to work and investigate the translational effect of successful WAs on the effectiveness of vocational outcomes, specifically RTW and, more importantly, long-term sustainability after RTW. The secondary aim is to consolidate and synthesize the quantitative and qualitative evidence in the literature, albeit separately. This review's final goal is to determine whether sex and gender variables are correlated with the effectiveness of accommodations in promoting re-entry into the workplace.

METHODS/DESIGN

PROSPERO

This protocol has been registered with PROSPERO (PROSPERO 2016:CRD42016043517), an international prospective register of systematic reviews database in health and social care, welfare, public health, education, crime, justice, and international development, where there is a health related outcome.

PATIENT AND PUBLIC INVOLVEMENT

No patient involved.

SEARCH STRATEGY

The search strategy was developed with an information specialist at the Toronto Rehabilitation Institute-University Health Network (TRI-UHN). A modified Population, Intervention, Comparison, Outcome (PICO) framework was used to facilitate and refine the search strategy in accordance with an evidence-based approach. The PICO framework was modified given that a comparison patient population and/or intervention was not considered. The search strategy was developed using a combination of comprehensive and specific subject headings and keywords, to balance sensitivity with specificity, and render articles across a wide spectrum of specific literature. The patient population was defined using broad terms such as "brain injuries", "craniocerebral trauma", "head injuries, closed" and "TBI" and the use of specific keywords which included but were not limited to "head", "intracran*", "concuss*" and "brain" combined with "trauma", "edema" and "lesion" using adjacent and wildcard operators. Broader terms such as 'return to work', "employment", "occupations" and "workplace" were used to search for articles, while specific narrow-focussed keywords such as "absenteeism", and "work" were searched with "status", "activit*", "abilit*" and "capacity" while additional keywords such as "return" were searched with "work" and "dut*" using the adjacent and wildcard operators to render relevant non-traditional publications. In the latter combination, the intend was to include a broad range of literature where the title, abstract and/or author provided keywords included traditional keywords like 'return to work' but also others variants such as 'returning back to work' or 'return to part time work' or 'returning to modified duties' or 'returned to preinjury duties' etc. Similarly, generic heading such as "equipment and supplies" and "Work Capacity Evaluation" were searched alongside "ergonomic", "jobsite" with "interven*", "duties" "counsel*, "training" and "approach*", again using the adjacent and wildcard operators. These alternative terms helped capture publications that may have been excluded with reliance on more commonly deployed workplace adaptation keywords.

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Search terms were also adapted from previously published systematic reviews on the topic [45 46]. The search terms and search strategy were independently peer-reviewed by another information specialist. Changes and suggestions were incorporated before the search terms were translated across nine databases, i.e., 1) Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations, 2) EBM Reviews Cochrane Central Register of Controlled Trials, 3) EBM Reviews Cochrane Database of Systematic Reviews, 4) Embase, 5) PsycINFO, 6) CINAHL, 7) Pubmed, 8) OT SEEKER, and 9) Health and Safety Science Abstracts. Additional supplementary file 1 provides the search strategy for each database. Furthermore, conference materials such as abstracts, editorials and/or commentaries will be excluded as they do not provide sufficient information required for the purposes of this review. All database results will be limited to English language. A limit to human only studies will be applied.

STUDY SELECTION

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for systematic reviews will be used to report the findings of this review [47].

Study selection will be completed by three reviewers, pre-determined eligibility criteria will guide all three reviewers and screening will be undertaken in two steps. Level I screen will involve screening of studies based on their titles and abstracts only. The Level I screen inclusion criteria are: 1) study participants must have sustained a TBI (irrespective of the mechanism or place of injury), 2) an outcome related to RTW must have been reported, and 3) study participants must be of working age (18-65 years). The primary outcome of interest, 'return-to-work', is defined as a post-injury attempt to resume competitive gainful employment. Each article may have one of three possible ratings, i.e., 'yes', 'no' and 'maybe'. All three reviewers will discuss articles that have non-matched ratings; if consensus is not achieved, mediation consultation will be undertaken. Articles that pass the first screen criteria will proceed to the full text Level II screen. The Level II screen entails a full-text screening of articles. Two additional criteria will be considered during the full-text Level II screen: 1) study participants were working at the time of the injury/loss, and 2) an intervention that includes WAs (of any kind) was provided to facilitate the RTW. The decision to screen for WAs during full-text screening is made after much deliberation as it is thought that a large proportion of titles and abstracts may not clarify the details of the interventions, specifically workplace accommodations, in an abstract when discussing return to work outcomes, hence, risking the erroneous elimination of articles that could have met the inclusion criteria.

For the purposes of this review, our operational definition of 'workplace accommodation' is "any measurable change and/or alteration in the process flow of an organization, materials used to accomplish a job demand, services provided/utilized at the workplace and/or routines for oneself (or others) to compensate for impairments or modify a work environment after a condition, disorder, injury and/or illness in an attempt to facilitate re-entry into the workforce" [Adapted from [48 49]]. All study designs will be included, for instance, randomized and non-randomized controlled trials, case-control studies, cohort and longitudinal studies, retrospective analysis, scoping or systematic reviews, case studies, qualitative studies and/or mixed methods research studies. Qualitative and case studies will particularly be included as they may contain details which quantitative exploration alone cannot capture. With respect to vocational status, all types of employment (full-time, part-time, casual, contract, self-employedetc.) and occupational categories will be considered. The same will be true for injury severity, whereby all mild,

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moderate and severe injuries will be considered. All types of injury mechanisms (blunt and penetrating trauma) will be included. Depending on the nature of the data, analysis will be stratified by sex, age and injury severity. Data from international publications, albeit written in English, will be included.

DATA EXTRACTION AND MANAGEMENT

All three reviewers will develop and pilot the data extraction form. The form will be modified to ensure that relevant data is captured for the assessment of study quality. The data extraction form will include author(s), study title, publication year, the year the study data was collected, purpose of study, study design, study setting (including the country of study, methods emloyed for recruiting study participants), sample size, demographic and vocational characteristics of the study participants, data source or type (e.g., retrospective, interviews etc.), severity of TBI and the diagnostic criterion utilized, statistical methodology utilized (e.g., descriptive, analytic and/or stratification techniques), comprehensive description of the intervention provided (e.g., employer-initiated, RTW co-ordinator, occupational therapist, case manager etc.), duration of RTW, subsequent follow-up(s), characteristics and costs of the accommodations provided, outcome of the stated intervention etc. Where possible, the outcome 'return to work' will be further clarified, i.e., distinctions will be made between return to work at the pre-injury job versus return to work at a new job, whether the new job is comparable to pre-existing job demands and/or remuneration and whether it is made possible with/without added training and/or accommodations and whether return to the pre-injury job occurred with modification and/or accommodation etc. If possible, secondary outcomes such as quality of life, health status and satisfaction indicators will also be reported in a separate table. The stated conflicts of interest and 'Risk of Bias' assessments will also be presented in a table.
Study data included after the level II screen will be extracted by two reviewers using a blinded extraction process. Reviewers will cross-review the collected data to document and address any discrepancies and rationales. Following this comparative analysis, the revised data extraction will be peer-reviewed by the third reviewer. Completed data extraction forms will be archived, and information from these forms will be entered and stored in Excel. One reviewer will independently check the data entry for accuracy.

QUALITY ASSESSMENT

The Critical Appraisal Skills Programme (CASP) checklists [50] will be used to assess study quality by guiding the assessment of validity and reliability of the included studies. The rationale for selecting the CASP checklists was made primarily due to the availability of separate checklists for different study designs such as cohort studies, review papers, randomized controlled trials and qualitative studies. Two reviewers will complete the checklists, through a blinded process, and later cross-review to document and address any discrepancies and rationales. Following consensus, the third reviewer will peer-review all checklists associated with this review's articles. With respect to missing data, study authors may be contacted for any queries regarding relevant data capture or additional data required to assess the risk of bias. Inter study bias will be reported separately in a table and may be discussed in the narrative.

DATA ANALYSIS, SYNTHESIS AND REPORTING

Data will be collected from each study and tabulated in a summary table. Due to the anticipated heterogeneity of study designs, patient cohorts, variability of outcome measures as

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well as personal and vocational factors (organizational structure, work industry, employment type etc.), outlining a single metric for workplace accommodations would undermine the richness of the existing data; hence, a meta-analytical approach or data pooling may not be feasible. Also, we anticipate that there will be few controlled studies that would be rated as high quality using typical level of evidence paradigms. Given the customizable and individualized nature of workplace accommodations, we believe that the richness of the data will actually come from small cohort, case control, case-series and qualitative studies. Below we have outlined our intended approach to incorporate findings from all levels of evidence.

Nevertheless, descriptive statistics will be utilized to characterize and synthesize the evidence from quantitative studies while data from qualitative studies will be summarized by identifying prominent and prevalent themes. A sequential exploratory mixed methods approach[51] will be utilized to analyze the extracted data. All relevant raw text narratives describing WAs provided to facilitate a RTW outcome will be compiled for each study. After reiterative readings, an inductive approach will be utilized to condense the relatively extensive raw data by assigning codes to the text narratives and establishing a consolidated group of broader meaningful categories for WAs. While a qualitative approach will be used to characterize and consolidate the data, a quantitative methodology will be used to determine statistically significant associations between WAs and the RTW outcome. This approach is intended to enhance the utility and impact of the findings [52]. A data analysis strategy [53] may be used to aggregate results, integrate the original meaning of the primary studies and reconstruct the findings to facilitate new interpretations of the data. The sources of the synthesized information from quantitative and qualitative studies will be reported separately.

Sub-group analysis: With sufficient data, sub-group analyses will be undertaken to investigate differences in outcomes. Given the expansion of the literature with respect to sex and gender differences in outcomes, two relevant Cochrane tools will be utilized, i.e., the Sex and Gender in Systematic Reviews Planning Tool and the Equity Checklist for Systematic Review Authors [54 55]. This unique investigation will clarify differences observed in data collection, analysis, and interpretation across studies. To examine the impact of interventions on inequality, the data will be further assessed using an intersectional lens to better understand the contributions of social determinants of health (e.g., age, race, education, socioeconomic status, etc.) on the outcomes of RTW and productivity.

The review will be undertaken according to this published protocol. Any deviations from the protocol will be reported in a separate sub-section and labeled "Differences between published protocol and review."

DISCUSSION

We believe this focused undertaking is the first of its kind. At this time, there have been no systematic reviews exploring the effectiveness of specific WAs that facilitate and/or sustain RTW after TBI. Reviews published to date have generally considered the combined role of care programs such as outpatient rehabilitation programs, generic workplace interventions, the impact of community-based remediation on RTW and/or return-to-productivity outcomes.

The Convention on the Rights of Persons with Disabilities (CRPD) [56], Disability Discrimination Act (DDA) [57] as well as the Canadian Human Rights Act (CHRC) [14] mandates equal rights and opportunities at the workplace, i.e., employees with disabilities are entitled to the same opportunities and benefits as people without disabilities. Under the legal test, CHRC,

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the employer has a duty to take reasonable steps to accommodate the position of an employee unless the employer's business would suffer 'undue hardship'. Although the Guides for Assessing Persons with Disabilities [49] have provided some guidance regarding the interpretation of undue hardship, operationalizing the construct of WAs has been short of easy. We believe this review will help shed light on the various conceptualizations of accommodations within the workplace and how WAs are operationalized within the practice of RTW following TBI. Investigating effective interventions that facilitate returning TBI survivors to work not only has the potential for short-term impact on a person's working life and ability to contribute to the economy, but also on the long-term health and well-being of society.

Given the evolving workforce, certain work-related supports are naturally embedded within the workplace, available for all employees and not considered a customized adjustment. These are often referred to as 'natural supports'. It is important to distinguish between WAs and natural supports, the former being a support customized for a particular worker who is encountering difficulties with work resumption. This review, however, will be focused specifically on WAs.

Effective WAs hold the potential to foster inclusive work designs for TBI survivors who are considering a return to employment. Given that a large proportion of TBI survivors face challenges reintegrating into the workforce, the study has implications from a labor market policy perspective. The information gathered and consolidated in this review will help health practitioners, policy makers, researchers and other workforce-related stakeholders make informed decisions when considering service delivery or future research into RTW for TBI survivors. With respect to service delivery, the 'low cost and high impact'[58] feature of WAs makes it not only easy to implement but also economically lucritive. Added benefits of WAs include retention of previously trained employees and hence elimination of costs associated with training a new employee, boosted morale amongst workers and an increased sense of well-being at both individual and societal levels.

Traditional RTW practices focus excessively on sickness and disability, whereby, based on a reported mismatch between the client's limitations and the demands of the job - a worker is deemed "unfit to work". In 2010, the United Kingdom replaced its "sickness certificate" with a statement of fitness for work also known as the "fit note" [59]. The shift in methodology from "not fit to work" to "fit to work" (if the employer is able to make necessary modifications) transformed the traditional "disability" lens into a "functional" focus. Currently, many procedural, legislative, training and practical issues remain regarding the nature of the ideal fit note, nevertheless, there is evidence to support that these notes, when filled correctly, are not only useful to the employer for planning an employee's return but are also improving RTW rates by more than 80% [60].In countries where workplace accommodations are legislated, health care practitioner's, physician's and employer's knowledge of effective WAs can facilitate work re-entry by ensuring an inclusive and safe work environment for the worker. The findings of this review may also be transferable across other sectors concerned with RTW or sustainability at work.

Effective WAs, for workers with special needs, may foster increased productivity which leads to economic well-being and eventually can be operationalized as the living standards for the population. Creating an inclusive labor market by removing barriers and providing employment opportunities to people with disabilities will not only generate a high impact return on the investment but also maximize the utility of human capital.

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DISSEMINATION PLAN

In addition to traditional knowledge translational approaches (i.e., publishing the findings in peer-reviewed journals, presenting at local and international conferences, posting summary-of-findings on university websites etc.), findings from this review will be disseminated using an integrated action-oriented knowledge translation framework [61]. It is hoped that this action-oriented framework will enable contextualization of the knowledge and findings to ensure impact in the real-world. Initial interest will be ignited through free webinars and podcasts as well as social media engagements, such as Twitter. Later, various stakeholders (i.e., Workplace Safety and Insurance Board, Insurance Bureau of Canada, Accessibilty Ontario etc.) will be invited to engage in a discussion to: 1) understand and contextualize the knowledge gained and 2) to explore how to reduce potential uptake barriers within the workplace environment. A special rehabilitation focus will be maintained to translate the knowledge to health practitioners, and to facilitate the development guidelines that will help facilitate the integration of these findings into day-to-day RTW practice.

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Contributors: AC and RS conceptualized the study. RS formulated the design, drafted the manuscript, assisted with the development of the search terms and registered the protocol via PROSPERO. AC provided expertise at each level and also reviewed the protocol. Both authors read and approved the final manuscript.

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References

1. Franulic A, Carbonell CG, Pinto P, et al. Psychosocial adjustment and employment outcome 2, 5 and 10 years after TBI. Brain Inj 2004;**18**(2):119-29.
2. Radford K, Phillips J, Drummond A, et al. Return to work after traumatic brain injury: cohort comparison and economic evaluation. Brain Inj 2013;**27**(5):507-20.
3. van Velzen JM, van Bennekom CA, Edelaar MJ, et al. How many people return to work after acquired brain injury?: a systematic review. Brain Inj 2009;**23**(6):473-88.
4. Sarajuuri JM, Kaipio ML, Koskinen SK, et al. Outcome of a comprehensive neurorehabilitation program for patients with traumatic brain injury. Arch Phys Med Rehabil 2005;**86**(12):2296-302.
5. Public Health Agency of Canada. Mapping Connections: An Understanding Of Neurological Conditions In Canada. 2014.
6. Centers for Disease Control and Prevention. TBI: Get the Facts 2019 [Available from: http://www.cdc.gov/traumaticbraininjury/get_the_facts.html.
7. National Center for Injury Prevention and Control. Report to Congress on Mild Traumatic Brain Injury in the United States: Steps to Prevent a Serious Public Health Problem. . 2003.
8. Cancelliere C, Kristman VL, Cassidy JD, et al. Systematic review of return to work after mild traumatic brain injury: results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch Phys Med Rehabil 2014;**95**(3 Suppl):S201-9.
9. Workplace Safety and Insurance Board. By The Numbers: 2014 WSIB Statistical Report. 2015 [Available from: Retrieved from: http://www.wsibstatistics.ca/.
10. Jantz PBC, Gail A. Child and adolescent traumatic brain injury: Academic, behavioural, and social consequences in the classroom. Support for Learning 2007;**22**(2):84-89.
11. Hendricks DJ, Sampson E, Rumrill P, et al. Activities and interim outcomes of a multi-site development project to promote cognitive support technology use and employment success among postsecondary students with traumatic brain injuries. NeuroRehabilitation 2015;**37**(3):449-58.
12. Artman L. K. MBT. Functional limitations in TBI and their relationship to job maintenance following work re-entry. Journal of Vocational Rehabilitation 39 2013 **39**:13–21.
13. Americans With Disabilities Act of 1990. Public Law 101-336.
14. Canadian Human Rights Act [R.S.C., c. H-6].
15. Mueller JL. Returning to work through job accommodation: a case study. AAOHN journal : official journal of the American Association of Occupational Health Nurses 1999;**47**(3):120-9; quiz 30-1.
16. Schultz IZ, Winter A, Wald J. Evidentiary support for best practices in job accommodation in mental health: macrosystem, employer-level and employee-level interventions. In: Schultz IZ, Rogers ES, eds. Handbook of work accommodation and retention in mental health. New York: Springer, 2010.
17. Targett P, Wehman P. Return to work after traumatic brain injury: a supported employment approach. In: Schultz IZ, ES R, eds. Handbook of work accommodatins and retention in mental health. New York: Springer, 2010.
18. Franche RL, Cullen K, Clarke J, et al. Workplace-based return-to-work interventions: a systematic review of the quantitative literature. J Occup Rehabil 2005;**15**(4):607-31.

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19. Franche RL, Severin CN, Hogg-Johnson S, et al. The impact of early workplace-based return-to-work strategies on work absence duration: a 6-month longitudinal study following an occupational musculoskeletal injury. J Occup Environ Med 2007;**49**(9):960-74.
20. Johnson KL, Yorkston KM, Klasner ER, et al. The cost and benefits of employment: a qualitative study of experiences of persons with multiple sclerosis. Arch Phys Med Rehabil 2004;**85**(2):201-9.
21. Gignac MA, Cao X, McAlpine J. Availability, need for, and use of work accommodations and benefits: are they related to employment outcomes in people with arthritis? Arthritis Care Res (Hoboken) 2015;**67**(6):855-64.
22. Wang J, Patten S, Currie S, et al. Perceived needs for and use of workplace accommodations by individuals with a depressive and/or anxiety disorder. J Occup Environ Med 2011;**53**(11):1268-72.
23. Franche RL, Severin CN, Hogg-Johnson S, et al. A multivariate analysis of factors associated with early offer and acceptance of a work accommodation following an occupational musculoskeletal injury. J Occup Environ Med 2009;**51**(8):969-83.
24. Target P, Wehman P, Petersen R, et al. Enhancing work outcome for three persons with traumatic brain injury. International journal of rehabilitation research Internationale Zeitschrift fur Rehabilitationsforschung Revue internationale de recherches de readaptation 1998;**21**(1):41-50.
25. Colantonio A, Salehi S, Kristman V, et al. Return to work after work-related traumatic brain injury. NeuroRehabilitation 2016;**39**(3):389-99.
26. Stergiou-Kita M, Mansfield E, Daiter L, et al. Good Intentions? Employer Representative Conceptualizations and, Challenges to the Workplace Accommodation Process: The Case of Electrical Injuries. Employ Respons Rights J 2015;**27**:1–25.
27. Stergiou-Kita M, Mansfield E, Colantonio A. Injured workers' perspectives on how workplace accommodations are conceptualized and delivered following electrical injuries. J Occup Rehabil 2014;**24**(2):173-88.
28. Saltychev M, Eskola M, Tenovuo O, et al. Return to work after traumatic brain injury: Systematic review. Brain Inj 2013;**27**(13-14):1516-27.
29. Kristman VL, Shaw WS, Reguly P, et al. Supervisor and Organizational Factors Associated with Supervisor Support of Job Accommodations for Low Back Injured Workers. J Occup Rehabil 2016.
30. Gagnon A, Lin J, Stergiou-Kita M. Family members facilitating community re-integration and return to productivity following traumatic brain injury - motivations, roles and challenges. Disabil Rehabil 2016;**38**(5):433-41.
31. Colantonio A. Sex, Gender, and Traumatic Brain Injury: A Commentary. Arch Phys Med Rehabil 2016;**97**(2 Suppl):S1-4.
32. Stergiou-Kita M, Mansfield E, Sokoloff S, et al. Gender Influences on Return to Work After Mild Traumatic Brain Injury. Arch Phys Med Rehabil 2016;**97**(2 Suppl):S40-5.
33. Keyser-Marcus LA, Bricout JC, Wehman P, et al. Acute predictors of return to employment after traumatic brain injury: a longitudinal follow-up. Arch Phys Med Rehabil 2002;**83**(5):635-41.
34. Kreutzer JS, Marwitz JH, Walker W, et al. Moderating factors in return to work and job stability after traumatic brain injury. J Head Trauma Rehabil 2003;**18**(2):128-38.

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35. Ownsworth T, McKenna K. Investigation of factors related to employment outcome following traumatic brain injury: a critical review and conceptual model. Disabil Rehabil 2004;**26**(13):765-83.
36. Crépeau F, Scherzer P. Predictors and indicators of work status after traumatic brain injury: A meta-analysis. . Neuropsychological Rehabilitation 1993;**3**(1):5-35.
37. Suominen S, Vahtera J, Korkeila K, et al. Job strain, life events, and sickness absence: a longitudinal cohort study in a random population sample. J Occup Environ Med 2007;**49**(9):990-6.
38. Cancelliere C, Donovan J, Cassidy JD. Is Sex an Indicator of Prognosis After Mild Traumatic Brain Injury: A Systematic Analysis of the Findings of the World Health Organization Collaborating Centre Task Force on Mild Traumatic Brain Injury and the International Collaboration on Mild Traumatic Brain Injury Prognosis. Arch Phys Med Rehabil 2016;**97**(2 Suppl):S5-18.
39. Kumar KS, Samuelkamaleshkumar S, Viswanathan A, et al. Cognitive rehabilitation for adults with traumatic brain injury to improve occupational outcomes. Cochrane Database Syst Rev 2017;**6**(6):Cd007935.
40. Awang H, Tan LY, Mansor N, et al. Factors related to successful return to work following multidisciplinary rehabilitation. J Rehabil Med 2017;**49**(6):520.
41. Dornonville de la Cour FL, Rasmussen MA, Foged EM, et al. Vocational Rehabilitation in Mild Traumatic Brain Injury: Supporting Return to Work and Daily Life Functioning. Front Neurol 2019;**10**:103.
42. Donker-Cools BHPM, Daams JG, Wind H, et al. Effective return-to-work interventions after acquired brain injury: A systematic review. Brain Injury 2016;**30**(2):113-31.
43. McDowell C, Fossey E. Workplace accommodations for people with mental illness: a scoping review. J Occup Rehabil 2015;**25**(1):197-206.
44. Alves DE, Nilsen W, Fure SCR, et al. What characterises work and workplaces that retain their employees following acquired brain injury? Systematic review. Occupational and environmental medicine 2020;**77**(2):122.
45. Nieuwenhuijsen K, Faber B, Verbeek JH, et al. Interventions to improve return to work in depressed people. Cochrane Database Syst Rev 2014(12):Cd006237.
46. van Vilsteren M, van Oostrom SH, de Vet HC, et al. Workplace interventions to prevent work disability in workers on sick leave. Cochrane Database Syst Rev 2015(10):Cd006955.
47. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Bmj 2009;**339**:b2535.
48. Job Accommodations Network. [Available from: http://askjan.org.
49. Public Service Commission of Canada. Guide for Assessing Persons with Disabilities: How to determine and implement assessment accommodations. 2007 [Available from: http://www.psc-cfp.gc.ca/plcy-pltq/guides/assessment-evaluation/apwd-eph/pdf/apwd-eph-eng.pdf.
50. Critical Appraisal Skills Programme (CASP) Checklists. 2013 [Available from: http://www.casp-uk.net/casp-tools-checklists.
51. Creswell JW, Plano Clark VL. Chapter 3: Choosing a Mixed Methods Design (pp. 53–106). Designing and Conducting Mixed Methods Research designs: Thousand Oaks, CA: Sage., 2011.

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52. Harden A. Mixed-Methods Systematic Reviews: Integrating Quantitative and Qualitative Findings.: A Publication of the National Center for the Dissemination of Disability Research (NCDDR). 2010.
53. Jang EE, McDougall DE, Pollon D, et al. Integrative Mixed Methods Data Analytic Strategies in Research on School Success in Challenging Circumstances. Journal of Mixed Methods Research 2008;**2**(3):221-47.
54. Doull M, Welch V, Puil L, et al. Development and evaluation of 'briefing notes' as a novel knowledge translation tool to aid the implementation of sex/gender analysis in systematic reviews: a pilot study. PLoS One 2014;**9**(11):e110786.
55. Ueffing E TP, Welch V, Petticrew M, Kristjansson E for the Campbell and Cochrane Equity Methods Group. Equity Checklist for Systematic Review Authors. 2012 [Available from: http://equity.cochrane.org/sites/equity.cochrane.org/files/uploads/EquityChecklist2012.pdf.
56. United Nations General Assembly. Convention on the Rights of Persons with Disabilities [Resolution 61/106]. [Available from: http://www.ohchr.org
57. Disability Discrimination Act. [1995, c. 50] [Available from: http://www.legislation.gov.uk/ukpga/1995/50/contents.
58. Office of Disability Employment Policy. Workplace Accommodations: Low Cost, High Impact. 2007 [Available from: http://digitalcommons.ilr.cornell.edu/gladnetcollect/383
59. Department for Work and Pensions. Guidance for GPs, employers, hospital doctors and occupational health practitioners on using fit notes (statements of fitness for work). 2013 [Available from: https://www.gov.uk/government/collections/fit-for-work-guidance
60. Coole C, Potgieter I, Nouri F, et al. Return-to-work outcomes and usefulness of actual fit notes received by employers. Family practice 2015;**32**(5):551-6.
61. Smith HA, McKeen JD, Singh S. Making knowledge work: five principles for action-oriented knowledge management. Knowledge Management Research & Practice 2006;**4**(2):116-24.

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Database: Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations and Ovid MEDLINE(R) <1946 to Present>
Search Strategy:
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1 Return to Work/ (929)

2 Employment, Supported/ (965)

3 Work/ (18878)

4 Employment/ (39799)

5 Unemployment/ (5713)

6 Occupations/ (21163)

7 Workplace/ (15989)

8 Occupational Medicine/ (23036)

9 Occupational Health/ (28115)

10 Occupational Health Services/ (10004)

11 Sick Leave/ (4524)

12 Absenteeism/ (7929)

13 (return* adj3 (work* or dut*)).tw,kw. (9843)

14 (employable or employee* or employment or unemployment or unemployed).tw,kw. (85298)

15 (sick leave* or sickness* or absence* or absenteeism).tw,kw. (538140)

16 ("disability management" or "disability prevention").tw,kw. (450)

17 (occupation or occupations or vocation or vocations).tw,kw. (31579)

18 (employability or employable or employee*).tw,kw. (36230)

19 ("work* adj2 abilit*" or ('work* adj2 capacity*') or ('work* adj2 activit*') or ('work adj2 disability*') or ('work* adj2 rehab*') or ('work* adj2 status*') or ('work* adj2 retention*') or (work* adj2 sustain*) or (work adj2 maintain*) or workabilit*).tw,kw. (22653)

20 or/1-19 (781782)

21 Disability Evaluation/ (39236)

22 exp Self-Help Devices/ (9342)

23 "Equipment and Supplies"/ (22368)

24 (case manager or case management).tw,kw. (9233)

25 (disab* adj3 guidance).tw,kw. (23)

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26 (assistive adj3 (device* or technolog* or self-help)).tw,kw. (2812)

27 (self help adj3 device*).tw,kw. (64)

28 Work Capacity Evaluation/ (5423)

29 Vocational Guidance/ (2040)

30 (vocational adj3 (training* or placement* or counseling)).tw,kw. (1818)

31 ((work* or job? or jobsite* or occupation* or employ*) adj5 interven*).tw,kw. (11855)

32 ((work* or job? or jobsite* or occupation* or employ*) adj5 accomod*).tw,kw. (22)

33 ((work* or job? or jobsite* or occupation* or employ*) adj5 modif*).tw,kw. (7043)

34 ((work* or job? or jobsite* or occupation* or employ*) adj5 adjust*).tw,kw. (5420)

35 ((work* or job? or jobsite* or occupation* or employ*) adj5 visit*).tw,kw. (3035)

36 ((work* or job? or jobsite* or occupation* or employ*) adj5 evaluat*).tw,kw. (32035)

37 ((work* or job? or jobsite* or occupation* or employ*) adj5 access*).tw,kw. (4463)

38 (reduced adj3 (employ* or occupation* or work* or duty or duties)).tw,kw. (3484)

39 (vocational adj3 guidance).tw,kw. (195)

40 (ergonomic adj3 (intervention? or counsel* or education or training or approach*)).tw,kw. (606)

41 ((light or modified or graded or alternative) adj3 (duty or duties or work*)).tw,kw. (4152)

42 or/21-41 (154976)

43 exp brain injuries/ (55732)

44 Craniocerebral Trauma/ (20483)

45 exp Head Injuries, Closed/ (8476)

46 exp Skull Fractures/ (19769)

47 mTBI*2.tw,kw. (1415)

48 tbi*2.tw,kw. (18022)

49 wrTBI*2.tw,kw. (8)

50 concuss*.tw,kw. (5252)

51 ((head* or cerebr* or crani* or capitis* or brain* or forebrain* or skull* or hemispher* or intracran* or orbit*) adj2 (injur* or trauma* or lesion* or damag* or wound* or destruction* or swell* or oedema* or edema* or fracture* or contusion* or commotion* or pressur*)).tw,kw. (158249)

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52 ((brain* or cerebr* or intracerebr* or crani* or intracran* or head* or subarachnoid* or subdural* or epidural* or extradural*) adj2 (haematoma* or hematoma* or hemorrhag* or haemorrhag* or pressur* or bleed*)).tw,kw. (76249)

53 or/43-52 (247310)

54 20 and 42 and 53 (440)

55 limit 54 to english language (369)

56 exp animals/ not (exp animals/ and exp humans/) (4244639)

57 55 not 56 (366)

58 (exp child/ or exp infant/) not ((exp child/ or exp infant/) and exp adult/) (1462717)

59 57 not 58 (361)

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Table 2| PRISMA-P (preferred reporting items for systematic review and meta-analysis protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Page No	Item No	Checklist item
Administrative information			
Title:			
Identification	1	1a	Identify the report as a protocol of a systematic review
Update	N/A	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	5	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:			
Contact	1	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	10	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	N/A	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:			
Sources	10	5a	Indicate sources of financial or other support for the review
Sponsor	10	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	10	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
Introduction:			
Rationale	3&4	6	Describe the rationale for the review in the context of what is already known
Objective	4	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
Methods:			
Eligibility criteria	5&6	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	5&6	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	Attached	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated

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Table 2| PRISMA-P (preferred reporting items for systematic review and meta-analysis protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Page No	Item No	Checklist item
Study records:			
Data management	6	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review
Selection process	5&6	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phaseof the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	6	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	6	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	6	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	6	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	7	15a	Describe criteria under which study data will be quantitatively synthesised
	7	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)
	7	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	7	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	6	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidenc	7	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

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